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Key Points:
- We assess the source-to-sink mineralogy of the Jezero crater paleolake system
- Alteration minerals in two fan deposits are detrital, not authigenic
- Aqueous alteration and paleolake activity are not genetically related

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Assessing the mineralogy of the watershed and fan deposits of the Jezero crater paleolake system, Mars

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Abstract We present results from geomorphic mapping and visible to near-infrared spectral analyses of the Jezero crater paleolake basin and its associated watershed. The goal of this study is to understand the provenance of the sedimentary deposits within this open-basin lake using a source-to-sink approach. Two fan deposits located within the basin have distinct visible to near-infrared mineralogic signatures measured by the Compact Reconnaissance Imaging Spectrometer for Mars (CRISM). The northern fan is spectrally characterized by a mixture of Mg-rich carbonate and olivine, while the western fan is characterized by Fe/Mg-smectite (e.g., saponite or nontronite) with variable amounts of Mg-rich carbonate and olivine in isolated exposures. The watersheds of these deposits contain a variety of geomorphic units that are likely to have supplied sediment to the Jezero crater paleolake, as the fluvial valleys that fed the basin incise these units. The geomorphic units include exposures of Fe/Mg-smectite-, olivine-, and Mg-rich carbonate-bearing terrain. We show that the difference in fan deposit mineralogy is a function of the areal exposure of the major geomorphic units within their watersheds. This indicates that the spectrally dominant aqueous alteration minerals in the fan deposits are primarily detrital, or transported, in nature and did not form in situ. We conclude that the aqueous alteration of the units in the watershed occurred prior to the fluvial activity that carved the valleys of the Jezero crater paleolake system, and that the two periods of aqueous activity are not genetically related.

1. Introduction

Over the past several decades, orbital and landed spacecraft data have provided an abundance of geomorphic evidence that indicates a period of widespread surface fluvial activity on Mars early in the history of the planet. The evidence includes the identification of long-inactive valley networks [e.g., *Pieri*, 1980; *Carr*, 1987; *Howard et al.,* 2005; *Irwin et al.,* 2005a, 2005b; *Fassett and Head,* 2008a; *Hynek et al.,* 2010] and paleolakes [e.g., *Goldspiel and Squyres*, 1991; *Cabrol and Grin,* 1999, 2001; *Fassett and Head,* 2005, 2008b; *Irwin et al.,* 2005a]. It is hypothesized that the majority of this fluvial activity peaked at approximately the Noachian-Hesperian boundary and ceased shortly thereafter [*Irwin et al.,* 2005a; *Fassett and Head,* 2008a; *Hoke and Hynek,* 2009; *Mangold et al.,* 2012].

Additionally, data acquired by the Observatoire pour la Minéralogie, l'Eau, les Glaces et l'Activité (OMEGA) instrument [*Bibring et al.,* 2004] on board the Mars Express (MEx) spacecraft and the Compact Reconnaissance Imaging Spectrometer for Mars (CRISM) instrument [*Murchie et al.,* 2007] on board the Mars Reconnaissance Orbiter (MRO) spacecraft have enabled widespread detections of hydrated alteration minerals, such as phyllosilicates and sulfates [e.g., *Gendrin et al.,* 2005; *Poulet et al.,* 2005; *Bibring et al.,* 2006; *Mustard et al.,* 2008; *Ehlmann et al.,* 2009, 2011; *Murchie et al.,* 2009a]. Identified phyllosilicate minerals are generally present in ancient, pre-Hesperian terrains [e.g., *Bibring et al.,* 2006; *Murchie et al.,* 2009a; *Ehlmann et al.,* 2011], lending support for early Martian aqueous activity.

While both morphologic and mineralogic evidence for ancient aqueous activity on Mars has been identified, the climatic conditions during this early period of Martian history are still poorly understood. Was the early Martian climate warmer and wetter than what we observe today with rainfall a common occurrence [e.g., *Craddock and Howard*, 2002], or was it as cold as modern Mars but icy (i.e., wetter than modern Mars) and with transient periods of warming that allowed for surface fluvial activity [e.g., *Wordsworth et al.,* 2013; *Halevy and Head,* 2014]? Studies of the geomorphology and inferred discharge estimates of Martian valley networks have hypothesized that liquid water precipitation (i.e., rainfall) is required for their formation [e.g., *Irwin et al.,* 2005a, 2005b; *Hoke and Hynek,* 2009; *Hynek et al.,* 2010], implying a Martian climate that is

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warmer and wetter than what is observed today; however, it has also been suggested that estimated discharges are consistent with surface runoff that would be produced by rapid melting of large amounts of snowpack [e.g., *Howard et al.,* 2005].

The identification of phyllosilicate minerals in ancient terrains on Mars has also been used to support the hypothesis that early Mars hosted a warmer, wetter surface environment [*Poulet et al.,* 2005; *Bibring et al.,* 2006; *Loizeau et al.,* 2007; *Carter et al.,* 2013], as many of these minerals form on Earth under pedogenic weathering conditions. However, recent work has suggested that the distribution of observed phyllosilicates, in particular Fe/Mg-rich smectites and/or mixed-layer clays, is more consistent with formation in the subsurface, without the need for warmer, wetter surface conditions for prolonged intervals [*Ehlmann et al.,* 2011]. In the former scenario, alteration minerals may have formed during the same interval of aqueous activity as the fluvial and lacustrine surface features, while in the latter scenario, the two occurred separately in space—alteration in the subsurface and fluvial activity at the surface—and perhaps also in time. Therefore, an important consideration for testing among these competing hypotheses is whether the period of fluvial activity on early Mars' surface is genetically linked to the formation of aqueous alteration minerals.

Paleolake deposits on Mars are an excellent geologic setting in which to investigate the relationship between alteration mineralogy and fluvial geomorphology. Many paleolakes on Mars exhibit both an inlet and outlet valley, requiring a standing body of water within the basin, and these features must have formed during a period of surface fluvial activity [*Fassett and Head,* 2008b]. A small number of these paleolakes have also been shown to contain lacustrine sedimentary deposits that are host to phyllosilicates and other aqueous alteration minerals [e.g., *Ehlmann et al.,* 2008a; *Dehouck et al.,* 2010; *Goudge et al.,* 2012a]. Understanding the regional stratigraphy of these paleolake systems can reveal the provenance of sedimentary alteration minerals, and whether they formed prior to or coincident with the fluvial activity associated with paleolake formation. Here we investigate the relationship between surface fluvial activity and aqueous alteration mineral formation for the Jezero crater paleolake system [*Fassett and Head,* 2005; *Schon et al.,* 2012] using a sediment mineralogy source-to-sink approach.

2. Background

2.1. Jezero Crater Geologic Context

Jezero crater is a ~45 km diameter impact crater located at 18.4°N, 77.7°E in the Nili Fossae region of Mars (Figure 1). The Nili Fossae are a series of approximately concentric graben located to the northwest of the Isidis basin, hypothesized to have formed as a tectonic response to the formation of the Isidis impact basin [*Comer et al.,* 1985; *Wichman and Schultz,* 1989]. The early history of this region was substantially shaped by the formation of the Isidis basin [*Mustard et al.,* 2007, 2009], an ~1900 km diameter impact crater [*Schultz and Frey,* 1990] that formed in the Middle-to-Early Noachian (~3.95–3.97 Ga) [*Werner,* 2008; *Fassett and Head,* 2011]. *Mustard et al.* [2007] point out that while the topography of the Isidis basin is not as well preserved as many large impact basins on the Moon, there still may be topographic remnants of the Isidis basin rings. This includes a chain of isolated peaks at a distance of ~1100 km from the basin center, one of which lies directly to the southeast of Jezero crater (Figure 1), which may be an inner ring of the basin [*Mustard et al.,* 2007].

The crustal material in the Nili Fossae region consists of both units emplaced prior to the Isidis basin forming event and Isidis-related materials, including extensive brecciated units and impact melt that are both well observed throughout the area [*Mustard et al.,* 2007, 2009]. The Nili Fossae region has also been subject to a variety of other geologic processes subsequent to the formation of the Isidis impact basin [e.g., *Ivanov et al.,* 2012], including fluvial activity and surface runoff [e.g., *Fassett and Head,* 2005; *Mangold et al.,* 2007], and volcanism, such as from the large, Hesperian-aged Syrtis Major volcanic complex [*Hiesinger and Head,* 2004].

The Nili Fossae region is also thought to be one of the locations on Mars with the largest diversity and most extensive exposures of alteration minerals [*Poulet et al.,* 2005; *Bibring et al.,* 2006; *Mustard et al.,* 2008; *Ehlmann et al.,* 2009], including exposures of Noachian crust that is altered to great depth [*Mangold et al.,* 2007; *Mustard et al.,* 2009; *Ehlmann et al.,* 2009, 2011]. *Mangold et al.* [2007] noted that Hargraves crater excavates alteration minerals exposed in the distal part of its ejecta, which they interpret to be sourced from the upper few hundred meters of the crust. *Mustard et al.* [2009] identify Fe/Mg-smectite in the fill of the Nili Fossae trough, again interpreted to represent deep (several hundred meters) alteration.

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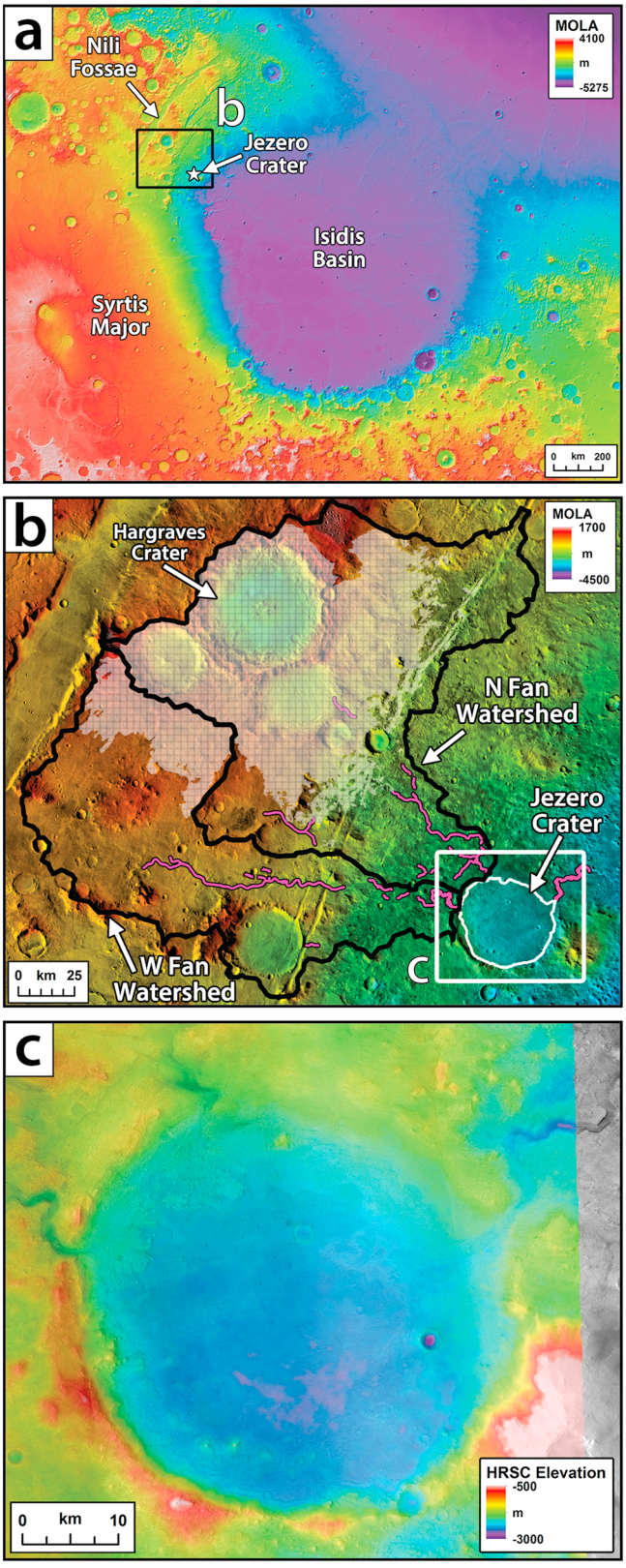

Saper and Mustard [2013] mapped >4000 m scale ridges in the Nili Fossae region, expanding on the work of *Head and Mustard* [2006], who identified ridges with a similar morphology exposed in the floors of eroded and exhumed impact craters. These ridges are widespread in the Nili Fossae region and are associated with aqueous alteration mineral-bearing units [*Saper and Mustard*, 2013]. *Head and Mustard* [2006] hypothesized that these ridges are crater-related faults and breccia dikes associated with the impact cratering process, while *Saper and Mustard* [2013] hypothesized that the ridges are impact-related fracture zones, possibly from the nearby Isidis basin, that have been subsequently mineralized. Widespread mineralized fracture zones in Nili Fossae are consistent with the hypothesis of pervasive alteration of the Noachian crust in this region of Mars [*Saper and Mustard,* 2013].

Figure 1. Jezero crater, Mars. North is up in all images. (a) Location map for Jezero crater (white star at 18.4°N, 77.7°E) in the Nili Fossae region of Mars. Syrtis Major, Nili Fossae, and the Isidis impact basin are also labeled. Location of blowup (Figure 1b) is indicated by labeled black box. Background is MOLA topography overlain on MOLA-derived hillshade. Image is centered at 13.65°N, 86.06° E. (b) Overview of the watershed areas for the northern and western fan deposits within the Jezero crater paleolake basin. Watershed areas are indicated in labeled, thick black outlines and were derived from modern MOLA topography. Mapped valley networks are indicated in magenta. The −2395 m MOLA contour, defined by *Fassett and Head* [2005] as the minimum level of ponded water within Jezero crater, is indicated in labeled, white outline. Regions within the watershed covered by material related to the Hargraves impact crater (e.g., ejecta, impact melt) are indicated in crosshatched area. Location of blowup (Figure 1c) is indicated by labeled white box. Background is MOLA topography overlain on the ~100 m/pixel THEMIS global daytime infrared mosaic. (c) Jezero crater paleolake basin. HRSC-derived DEM h5270_0000 overlain on a mosaic of CTX images P04_002664_1988, P06_003376_1987, P06_003521_1971, P15_007068_1971, and P17_007714_2001.

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2.2. Previous Work on the Jezero Crater Paleolake

Jezero crater has two inlet valleys, one on the northern rim and one on the western rim, and an outlet valley that breaches the eastern crater rim (Figure 1). This requires that the basin contained a standing body of water and was a hydrologically open lake at one point in time [*Fassett and Head,* 2005, 2008b; *Schon et al.,* 2012]. Evidence for a paleolake within Jezero crater was first described by *Fassett and Head* [2005], who identified the inlet and outlet valleys feeding and draining a basin enclosed by a topographic contour ~250 m above the current crater floor. This contour is at an elevation of −2395 m relative to the Mars datum and defines the minimum level of ponding within the crater prior to breaching to form the outlet valley [*Fassett and Head,* 2005]. *Fassett and Head* [2005] also identified two distributary fan deposits at the mouth of each inlet valley.

The duration of the Jezero crater paleolake system was studied in detail by *Schon et al.* [2012]. The authors provide detailed observations of meander and scroll bar migration within the western Jezero fan deposit, which they argue could have formed in ~20–40 years based on terrestrial scroll bar migration rates. Additionally, the amount of infill within Jezero crater compared to a similar sized, fresh impact crater (~1 km of fill) was used to estimate a time scale of activity of ~10^6–10^7 years. This estimate is based on terrestrial lake sediment accumulation rates, and the assumption that the entire thickness of fill is lacustrine in origin [*Schon et al.,* 2012].

Fassett and Head [2008a] performed buffered crater counts on the valleys that fed and drained Jezero crater, and found an age of cessation for the fluvial system of ~3.8 Ga using the Neukum production function [*Ivanov,* 2001], during the Late Noachian based on the absolute age boundaries of *Werner and Tanaka* [2011]. This age is broadly consistent with the age of cessation for 25 other valley network systems dated by *Fassett and Head* [2008a], as well as other estimates of the end of valley network formation [e.g., *Irwin et al.,* 2005a; *Hoke and Hynek,* 2009; *Mangold et al.,* 2012].

The age of lacustrine activity within Jezero crater is also constrained by a volcanic unit that is present on the floor of the crater and embays both fan deposits [*Goudge et al.,* 2012b; *Schon et al.,* 2012]. *Goudge et al.* [2012b] derive an emplacement age of this unit of ~3.45 Ga using the Neukum production function of *Ivanov* [2001], during the Early Amazonian based on the absolute age boundaries of *Werner and Tanaka* [2011]. *Schon et al.* [2012] also note that the fan deposits were eroded from their original extent prior to being embayed by the volcanic unit, and that the fans have been preferentially eroded compared to the volcanic floor unit subsequent to the emplacement of this unit.

Ehlmann et al. [2008a, 2008b, 2009] used CRISM data to identify Fe/Mg-smectite and Mg-rich carbonate within the Jezero crater fan deposits. It is hypothesized that these sedimentary alteration minerals are transported (i.e., detrital) in nature as opposed to having formed in situ (i.e., authigenic) [*Ehlmann et al.,* 2008a, 2008b, 2009], due to the identification of both Fe/Mg-smectite and Mg-rich carbonate in the surrounding Nili Fossae region [*Mangold et al.,* 2007; *Mustard et al.,* 2008, 2009; *Ehlmann et al.,* 2008a, 2008b, 2009]. However, this hypothesis has yet to be rigorously tested with a source-to-sink mineralogy analysis of the Jezero crater watershed and paleolake basin, including the fan deposits. The work presented here is aimed at directly testing this hypothesis to further understand the Jezero crater paleolake system, as well as the relationship between fluvial activity and aqueous alteration on early Mars.

3. Data Used and Methods

Our Jezero crater paleolake system source-to-sink analysis employs a combined approach of geomorphic mapping and mineralogic analysis using visible to near-infrared (VNIR) hyperspectral reflectance data. We first mapped the major geomorphic units within the Jezero basin and surrounding watershed area to establish an understanding of the major regional units and the overall stratigraphy of these units. Following this, we studied the mineralogy of the major units in the basin as well as the units in the watershed that are stratigraphically below the fluvial activity, i.e., those units that could have acted as source material for the sedimentary deposits within Jezero crater. We then compared the mineralogy of the paleolacustrine sedimentary deposits in Jezero crater with the mineralogy of the major geomorphic units in the watershed to assess the source-to-sink mineralogy of the paleolake system.

3.1. Geomorphic Mapping

The primary image data used are from the Context Camera (CTX) [*Malin et al.,* 2007] and High Resolution Imaging Science Experiment (HiRISE) [*McEwen et al.,* 2007] instruments on board the MRO spacecraft. CTX

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Table 1. CTX Images Used in Producing Regional Mosaic for Geomorphic Mapping

Image ID Number	Resolution (m/pixel)
B02_010272_2009	6.55
B02_010496_2005	5.65
B03_010773_2000	5.57
B06_011841_2005	5.69
B11_013964_2038	5.62
B12_014386_1994	5.56
P04_002664_1988	6.92
P05_002809_1975	5.53
P05_003086_2011	5.69
P06_003376_1987	5.55
P06_003521_1971	5.62
P07_003587_2005	6.41
P08_004009_1982	5.55
P13_006211_2038	5.64
P13_006277_1982	5.55
P14_006488_2006	5.55
P15_006712_2010	5.7
P15_006778_2002	5.64
P15_006844_2009	5.63
P15_007068_1971	5.54
P17_007714_2001	5.61
P17_007780_1987	5.65
P17_007846_2007	5.62
P19_008281_1982	5.55
P19_008426_2022	5.8
P20_008716_1998	6.44
P21_009151_2014	5.75

provides images at ~6 m/pixel and has good global coverage, offering an appropriate data set for regional geomorphic mapping [*Malin et al.,* 2007]. For use in our mapping, we created a mosaic of 27 CTX images (Table 1) that covers the entire Jezero crater basin and watershed area except for a ~2 × 20 km image gap in the western part of the watershed. The CTX image mosaic was produced at a resolution of ~7 m/pixel with the U.S. Geological Survey Integrated Software for Imagers and Spectrometers (ISIS). HiRISE images at a spatial scale of ~0.25 m/pixel were used for detailed geomorphic analysis with more limited spatial coverage [*McEwen et al.,* 2007]; at the time of writing, there are 78 available HiRISE images that are within or cover some portion of our map area, although only select images were used in our analysis. In addition, we used the ~100 m/pixel global daytime infrared mosaic [*Edwards et al.,* 2011] from the Thermal Emission Imaging System (THEMIS) instrument on board the Mars Odyssey spacecraft [*Christensen et al.,* 2004].

The main topographic data used in this work are from the Mars Orbiter Laser Altimeter (MOLA) instrument on board the Mars Global Surveyor spacecraft [*Smith et al.,* 2001] and the High Resolution Stereo Camera (HRSC) instrument on board the MEx spacecraft [*Neukum et al.,* 2004; *Gwinner et al.,* 2010]. We used MOLA-gridded topography at ~460 m/pixel, which provides nearly complete global topographic information [*Smith et al.,* 2001]. We used HRSC stereo-derived digital elevation models (DEMs) h0988_0000 (75 m/pixel), h1347_0000 (100 m/pixel), h3047_0000 (50 m/pixel), and h5270_0000 (75 m/pixel), which cover the Jezero basin and watershed area. These DEMs were mosaicked at a resolution of ~100 m/pixel using ESRI's ArcMap geographic information system (GIS) software. For detailed topographic information, we also used stereo-derived DEMs produced from images acquired by HiRISE using the NASA Ames Stereo Pipeline [*Broxton and Edwards,* 2008; *Moratto et al.,* 2010]. To ensure accurate topographic measurements, the HiRISE DEMs were tied (i.e., rotated and translated in three dimensional space to reduce the error between two topographic data sets) to the HRSC DEM mosaic, which was also tied to MOLA point shot (i.e., nongridded) topography, using the NASA Ames Stereo Pipeline [*Beyer et al.,* 2014].

All of the data were georeferenced and coregistered in ESRI's ArcMap GIS software. We delineated the watersheds of the inlet valleys associated with the two Jezero fan deposits [*Fassett and Head,* 2005; *Schon et al.,* 2012] using MOLA-gridded topography along with the ESRI ArcMap Hydrology Toolset [*ESRI,* 2012]. Here we refer to the watershed of each of the inlet valleys as the area, defined by topography, from which all surface runoff will eventually flow to the point where the inlet valley debouches into the Jezero crater paleolake, i.e., the inlet point. The inlet point for each of these watersheds was chosen as the point where the inlet valley intersects the −2395 m MOLA contour within Jezero crater [*Fassett and Head*, 2005]. This resulted in a calculated watershed for each of the two fan deposits based on current MOLA topography (Figure 1b). The two watersheds were then manually examined and slightly modified based on analysis with the HRSC DEM and CTX image mosaics.

The Jezero crater watershed calculated here has an area of ~30,700 km^2 (~12,000 km^2 for the western fan, and ~18,700 km^2 for the northern fan). This is approximately twice as large as the manually mapped watershed of

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Table 2. CRISM Observations Analyzed in This Study

CRISM Observation ID[a]	Resolution (m/pixel)	Relevant Units[b]
FRT0000289E	18.1	PCu, Bdm
FRT000047A3	18.8	Fn, Fw, LTF
FRT00005850	24.2	LTF, MT, Crw
FRT000066A4	18.8	VF, Fw, LTF
FRT0000722C	18.2	Ce, Crw
FRT000093BE	18.7	Cim, MTe, Bal, Balr, Bdm
FRT000097E2	18.8	Cim, MTe, Bal, Balr, Bdm
FRT0001182A	18.3	MT, Bdm
FRT00013EBC	18.8	PCu, PCul, Bdm
FRT0001C558	18.3	MT, Crw
FRT0001EAE0	18.7	VF, LTF
FRT0001ECBA	18.9	VF, LTF
FRT0001FB74	19.4	VF, LTF, MT
FRT00021DA6	18.6	VF, LTF
FRT00023370	19.8	Cim, PCu, MTe, Balr
HRL000040FF	36.7	VF, Fn, Fw, LTF, MT
HRL000095A2	36.8	Bdm, PCu
HRL00010963	37.0	Crw, MTl, Bdm
HRL00010D72	37.4	MTe, Balr
HRL000116C6	37.3	Crw, MTl, Bdm

[a]See Figure 2 for CRISM image locations.
[b]Relevant units are those contained within the image, listed by unit map symbol from Tables 3 and 4.

Fassett and Head [2005], which has a calculated area of ~15,000 km^2. This discrepancy is likely due to automated versus manual identification of the watershed boundaries, and in particular the area affected by the formation of the Hargraves impact crater (Figure 1b). This area occupies a substantial portion of the watershed delineated from our automated technique; however, *Fassett and Head* [2005] did not include much of this area in their manually mapped watershed, as the Hargraves impact crater superposes the Jezero inlet valley networks. The formation of the Hargraves impact crater substantially modified the topography of the Jezero crater paleolake system, making it difficult to accurately determine the true watershed area for this basin. It is, therefore, possible that our watershed area is an overestimate of the Jezero crater paleolake watershed; however, for geomorphic mapping purposes, we consider this to be a reasonable approach, as mapping a larger area will typically reveal more unit contacts and stratigraphic relationships, providing more information with which to define the regional stratigraphy.

Given these watershed areas (Figure 1b), we mapped the observable geomorphic units within: (1) the two watersheds, (2) the Jezero basin, defined by the −2395 m MOLA contour [*Fassett and Head,* 2005], and (3) a ~70 × 70 km region centered on the basin. The geomorphic map was completed at a scale of 1:100,000, with mapping done at a scale of ~1:30,000 using the CTX mosaic (Table 1). Geomorphic units were defined based on tone, texture, and morphologic characteristics (e.g., tendency of the unit to form mesas or pits). Unit definitions were based on consistently observed geomorphic characteristics in the mapping area based on the CTX image mosaic, as well as supporting HiRISE images, the THEMIS global daytime infrared mosaic, and the HRSC DEM mosaic. Unit boundaries were mapped based on observable contacts between geomorphic units with distinct defining characteristics; however, in some locations the contact between distinct geomorphic units is either obscured or gradational, in which case unit boundaries were approximated. The stratigraphic position and crosscutting relationships of the defined geomorphic units relative to one another were assessed using both the CTX image mosaic and the HRSC DEM mosaic.

3.2. Spectral Analysis

The mineralogy of the units identified through geomorphic mapping was analyzed using data from the CRISM instrument. CRISM is a VNIR imaging spectrometer on board MRO that measures reflected light from the surface of Mars across the wavelength range ~0.36–3.9 μm. Here we focus on full spectral resolution (spectral sampling of ~6.5 nm/channel) and full (~18 m/pixel; full-resolution targeted, FRT) and half (~36 m/pixel; half-resolution long, HRL) spatial resolution targeted images [*Murchie et al.,* 2007]. These data enable an examination of the best exposures of the mapped geomorphic units in order to extract spectra least affected by atmospheric and surface dust effects, which give the best indication of the spectrally dominant mineralogy. We analyzed a total of 20 CRISM targeted image cubes in the mapping area (Table 2), which are primarily concentrated in the Jezero crater basin and northern fan watershed (Figure 2).

CRISM data were converted from radiance measured by the sensor to I/F through division by a solar spectrum [*Murchie et al.,* 2007, 2009b]. The calibrated I/F data were downloaded from the Planetary Data System (PDS), corrected for photometry (i.e., cosine of the incidence angle), and atmospherically corrected using a volcano scan correction [e.g., *Mustard et al.,* 2008; *Ehlmann et al.,* 2009; *McGuire et al.,* 2009; *Murchie et al.,* 2009a, 2009b].

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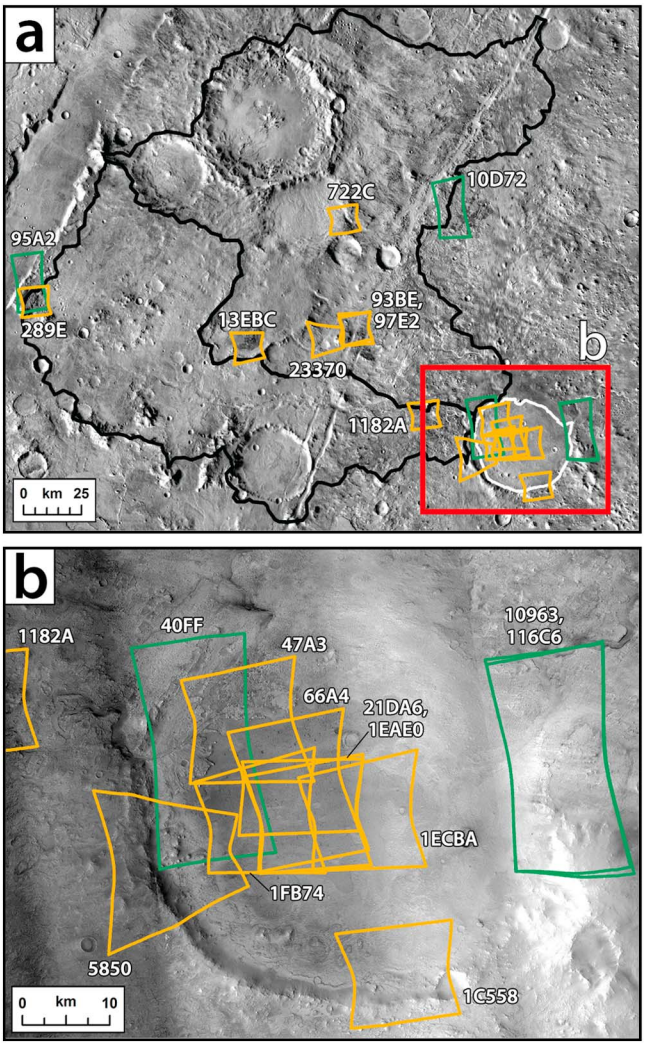

Figure 2. Locations of CRISM images analyzed in this work. Orange outlines indicate full-resolution targeted (FRT) images, and green outlines indicate half-resolution long (HRL) images. Images are labeled with four or five character hexadecimal IDs. North is up in all images. (a) Analyzed CRISM images within the Jezero crater paleolake watershed. Thick black lines indicate the northern and western fan deposit watersheds, and white line indicates the outline of the Jezero crater paleolake basin (−2395 m MOLA contour). Location of blowup (Figure 2b) is indicated by labeled red box. Background is the ~100 m/pixel THEMIS global daytime infrared mosaic. (b) Analyzed CRISM images within and around Jezero crater. Background is a mosaic of CTX images P04_002664_1988, P06_003376_1987, P06_003521_1971, P15_007068_1971, and P17_007714_2001.

The volcano scan correction operates under the assumption that atmospheric and surface contributions to the measured signal are multiplicative, and that the strength of atmospheric absorptions varies with the path length through the atmosphere following a power law [*Bibring et al.,* 1989]. A two-way atmospheric transmission spectrum is estimated by taking a ratio of a spectrum taken at the summit of Olympus Mons and a spectrum from lower on the flank of the volcano. This atmospheric transmission spectrum is then scaled for the observation of interest to account for differences in the atmospheric pathlength due to differences in elevation and removed from each observation via division to approximate the surface reflection [e.g., *Mustard et al.,* 2008; *Ehlmann et al.,* 2009; *McGuire et al.,* 2009; *Murchie et al.,* 2009a, 2009b].

Spectral parameter maps [*Pelkey et al.,* 2007; *Ehlmann et al.,* 2009; *Salvatore et al.,* 2010; *Viviano-Beck et al.,* 2014] were calculated from the atmospherically corrected CRISM data, and these maps were used as guides for manual spectral analysis to determine the mineralogy of the mapped geomorphic units. The corrected CRISM data, along with the spectral parameter maps, were geographically projected [*Murchie et al.,* 2007, 2009b] and analyzed alongside the coregistered CTX mosaic and geomorphic map.

In an attempt to reduce noise and extract the best possible spectra from the identified geomorphic units, large (typically on the order of hundreds of pixels) regions of interest (ROIs) were used to calculate average spectral signatures. These average spectra were ratioed to ROI spectra of nearby, spectrally bland terrain in an attempt to remove any residual instrument noise and bring out the unique spectral characteristics of the unit of interest [e.g., *Mustard et al.,* 2008; *Ehlmann et al.,* 2009]. Care was taken to use denominator spectra that have minimal spectral features, so as not to introduce anomalous spectral shapes or features in the ratioed spectrum. We have also used only denominator spectra that are from the same columns as the average ROI spectrum of the unit of interest to reduce the effect of column-dependent instrument noise [*Murchie et al.,* 2007, 2009b]. Using these techniques, representative spectra from the major geomorphic units were collected from multiple locations within the watershed and within the Jezero basin itself.

Spectra extracted from the major geomorphic units in the mapping area were characterized based on the presence, shape, and position of major absorption features in ratioed I/F spectra, and compared to a suite of

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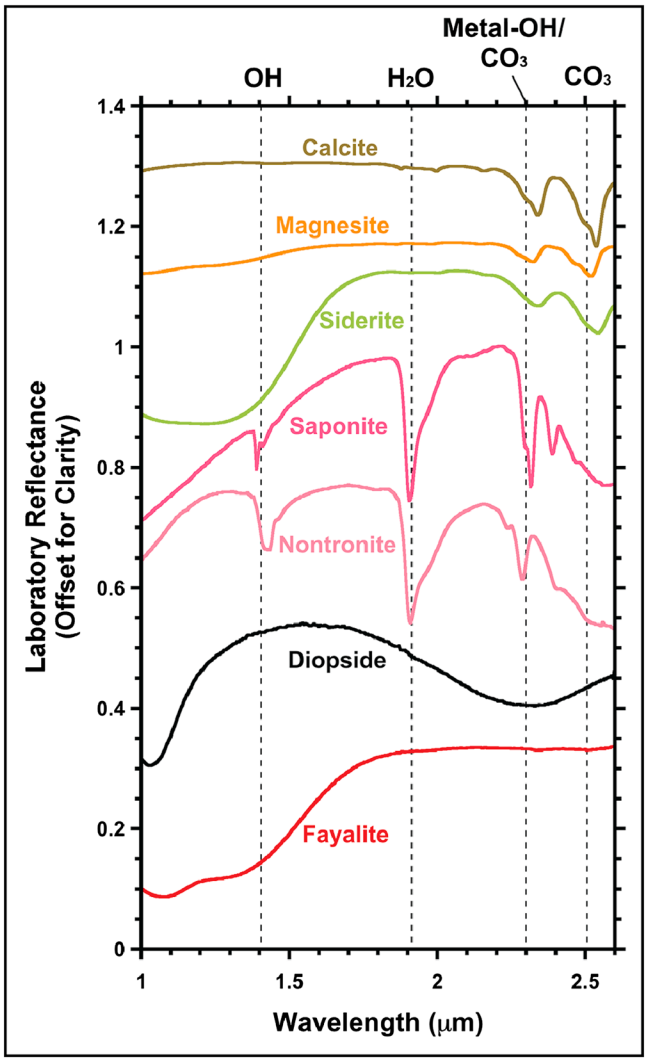

Figure 3. Example laboratory reflectance spectra relevant to the spectral signatures of the geomorphic units within the Jezero crater paleolake basin and watershed. Spectra are from the CRISM spectral library [*CRISM Science Team*, 2006]. From top to bottom, spectra are samples KACB09A (calcite), KACB03A (magnesite), KACB08B (siderite), LASA53 (saponite), CCJB26 (nontronite), LAPP97 (diopside), and C1PO58 (fayalite). Dashed lines are located at ~1.41, 1.92, 2.3, and 2.51 μm.

laboratory-measured mineral spectra (e.g., Figure 3). The main spectrally distinct minerals previously identified in the Jezero crater paleolake and surrounding terrain include Fe/Mg-smectite, Mg-rich carbonate, olivine, and pyroxene [*Mangold et al.,* 2007; *Mustard et al.,* 2007, 2009; *Ehlmann et al.,* 2008a, 2008b, 2009], which all exhibit diagnostic vibrational and crystal field absorptions (Figure 3).

Laboratory spectra of Fe/Mg-smectite, such as the Fe-rich nontronite or Mg-rich saponite, display prominent vibrational absorptions at ~1.4, 1.9, and 2.29–2.31 μm [*Clark et al.,* 1990; *Bishop et al.,* 1999, 2002; *Frost et al.,* 2002] (Figure 3). Specifically, the absorption at ~1.4 μm is caused by the first stretching overtone of OH in the mineral structure as well as a combination tone from structural H_2O, the absorption at ~1.9 μm is caused by a combination tone of OH bend and H–O–H stretch in structural H_2O, and the ~2.29–2.31 μm absorption is caused by a combination tone of OH stretch and Fe/Mg-OH bend in the smectite structure [*Clark et al.,* 1990; *Bishop et al.,* 1999, 2002; *Frost et al.,* 2002]. Fe-rich smectites such as nontronite have metal-OH bands centered near 2.29 μm, while Mg-rich smectites such as saponite have metal-OH bands centered near 2.30–2.31 μm [*Clark et al.,* 1990; *Bishop et al.,* 1999, 2002; *Frost et al.,* 2002]; however, band center determination at the ~10 nm level is difficult with CRISM data, and thus we refer to similar spectral signatures as Fe/Mg-smectite throughout this paper.

Laboratory carbonate spectra have two sets of prominent VNIR spectral features: (1) a set of sharp, paired absorptions at ~2.3 and 2.5 μm, and (2) stronger, broader absorptions at ~3.4 and 3.9 μm. The paired absorptions at ~2.3 and 2.5 μm are caused by an overtone of a fundamental vibrational mode of CO_3 in the carbonate mineral structure, with the precise position of these absorptions controlled by the dominant cation (e.g., Mg for magnesite, Ca for calcite, Fe for siderite) [*Hunt and Salisbury,* 1971; *Gaffey,* 1987] (Figure 3). The ~3.4 and 3.9 μm absorptions in carbonate are related to combination tones and overtones of fundamental vibrational modes of CO_3 in carbonate minerals [*Schroeder et al.,* 1962]; however, the presence of water, either in hydrous carbonates or in mixtures of anhydrous carbonate and hydrous minerals, can act to suppress the identification of these absorption features [e.g., *Calvin et al.,* 1994]. Carbonate minerals on Mars are typically identified by paired absorptions at ~2.3 and 2.5 μm [*Ehlmann et al.,* 2008b, 2009].

Both olivine and pyroxene laboratory spectra display broad absorption features that are caused by electronic crystal field transitions from octahedrally coordinated Fe^{2+} in their mineral structure [*Adams,* 1974; *King and Ridley,* 1987; *Burns*, 1993] (Figure 3). Olivine spectra have a prominent, broad absorption feature, centered

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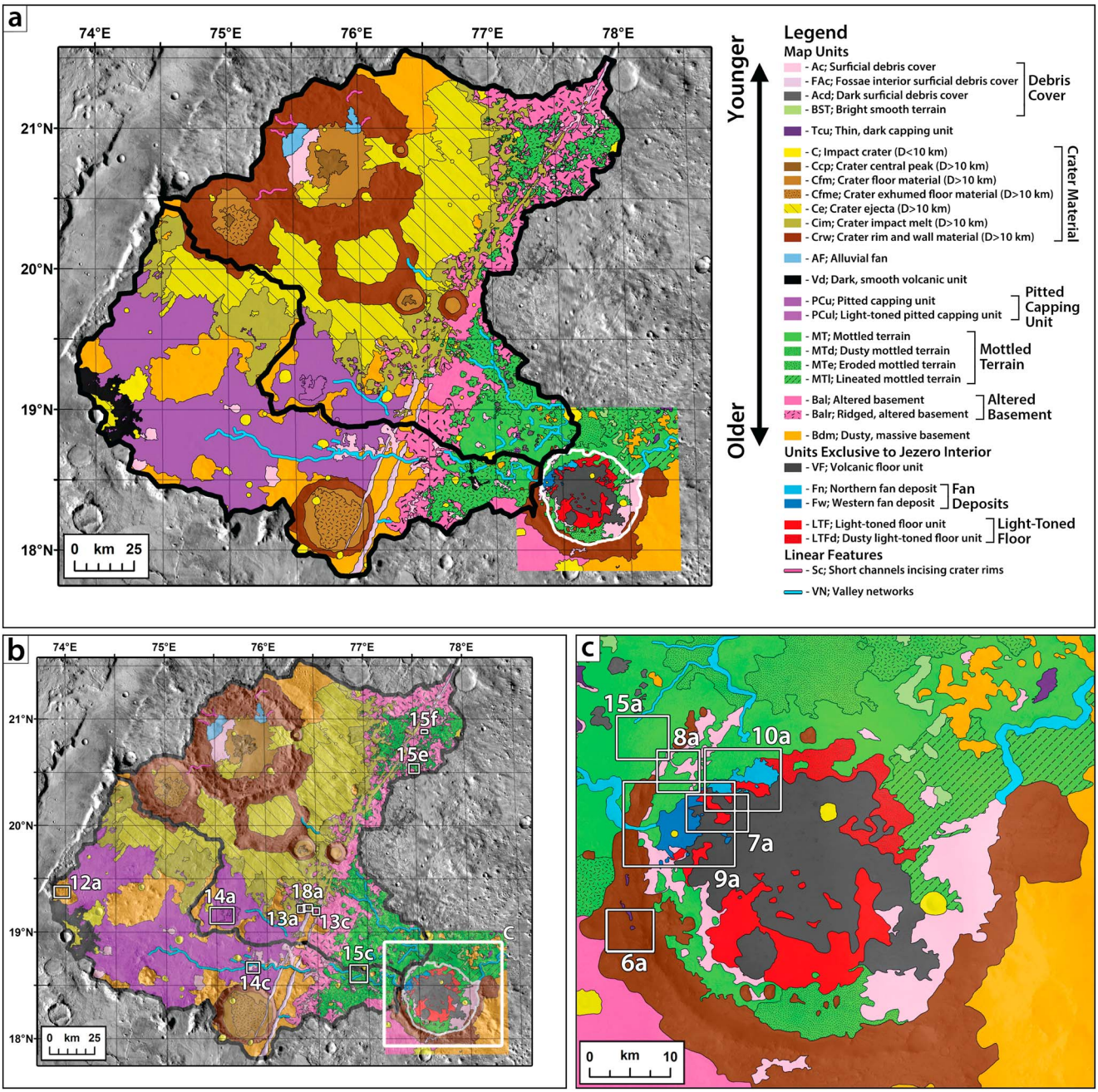

Figure 4. (a) Geomorphic map of the Jezero crater paleolake basin, surrounding area, and watershed. Legend is shown on the right, with units listed in approximate stratigraphic order. Units found exclusively within Jezero crater are listed separately, also in approximate stratigraphic order. See main text and Tables 3 and 4 for description of units. Thick black lines indicate the northern and western fan deposit watersheds, and white line indicates the outline of the Jezero crater paleolake basin (−2395 m MOLA contour). Background is the ~100 m/pixel THEMIS global daytime infrared mosaic. (b) As in Figure 4a but with locations indicated for geomorphic example figures shown throughout the paper. Location of blowup (Figure 4c) is indicated by labeled white box. (c) Geomorphic map focused on the Jezero crater paleolake basin. Locations indicated for geomorphic example figures shown throughout the paper. Background is a mosaic of CTX images P04_002664_1988, P06_003376_1987, P06_003521_1971, P15_007068_1971, and P17_007714_2001. North is up in all images.

near ~1 μm, that is a composite of three separate absorptions with the precise band center and shape controlled by the Fe^{2+} content of the olivine [*King and Ridley,* 1987; *Burns*, 1993] (Figure 3). Pyroxene spectra display two broad absorption features centered near ~1 and 2 μm, with the precise position of these absorptions primarily controlled by the Ca content of the pyroxene—higher Ca content shifts the ~1 and 2 μm band centers to longer wavelengths [*Adams,* 1974; *Burns,* 1993; *Klima et al.,* 2007, 2011].

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Table 3. Geomorphic Units Mapped Within the Jezero Crater Paleolake Watershed[a]

Unit Map Symbol	Unit Name	Unit Description	Unit Interpretation	Relevant Figure(s)
Ac	Surficial debris cover	Small patches of surficial cover that are typically aggregated into dunes, although can also appear as mass wasting or general dust cover. Always the most surficial deposit observed.	Surficial debris, mobilized and emplaced by aeolian activity.	8–10, 13c, 13d, 14 c, and 18a
FAc	Fossae interior surficial debris cover	Similar morphology as the surficial debris cover unit (Ac), but contained within the Nili Fossae interior.	Same unit as surficial debris cover (Ac) within the Nili Fossae interior.	18a
Acd	Dark surficial debris cover	Dark toned surficial deposits that are most commonly found in local topographic lows, and often aggregate into dunes.	Darker surficial debris cover (Ac).	
BST	Bright smooth terrain	Small areas of bright, smooth terrain, possibly with observable dunes at the finest scales. Found only in local topographic lows within the mottled terrain unit (MT).	Surficial debris cover (Ac) eroded from the mottled terrain unit (MT) into fine material filling local topographic lows.	
Tcu	Thin, dark capping unit	Small patches of a smooth, thin, cliff-forming, dark toned unit that sits stratigraphically above whatever unit it is emplaced upon.	Capping unit with unknown origin.	
C	Impact crater ($D < 10$ km)	Crater ejecta, rims, and floor material of all craters with a diameter < 10 km.	Impact crater and associated deposits (e.g., ejecta, rim material, etc.) with a diameter < 10 km.	9, 14a, 14b, 15c, and 15d
Ccp	Crater central peak ($D > 10$ km)	Blocky, uplifted central peak material within craters with a diameter > 10 km.	Central peaks of craters with a diameter > 10 km.	
Cfm	Crater floor material ($D > 10$ km)	All material associated with floors of craters with a diameter > 10 km.	Floor material of craters with a diameter > 10 km.	
Cfme	*Crater exhumed floor material ($D > 10$ km)*	*Rough, textured material that appears eroded and exhumed on the floors of craters with a diameter > 10 km.*	*Floor material of craters with a diameter > 10 km that has been buried and subsequently exhumed.*	
Ce	Crater ejecta ($D > 10$ km)	Rough ejecta material that shows radial textures from the crater that emplaced it. Often intermixed with small patches of impact melt (Cim). Only mapped for craters with a diameter > 10 km.	Ejecta of craters with a diameter > 10 km. Deposits may also have small patches of impact melt (Cim) interspersed, but the two cannot be easily separated at the mapping scale.	
Cim	Crater impact melt ($D > 10$ km)	Dark to moderate toned, smooth, thin, coherent unit. Often shows polygonal cracks at the margins in HiRISE images. Material has clearly flowed around high-standing topography. Only mapped for craters with a diameter > 10 km.	Distal impact melt of craters with a diameter > 10 km.	13a, 13b, and 18
Crw	Crater rim and wall material ($D > 10$ km)	Blocky rim and wall material associated with impact craters with a diameter > 10 km.	Rim and wall materials of craters with a diameter > 10 km.	6 and 9
AF	Alluvial fan	Depositional fan deposits on the interior walls of the Hargraves impact crater.	Alluvial fans emplaced by fluvial activity along the rim of the Hargraves impact crater [*Mangold et al.,* 2007].	
Vd	Dark, smooth volcanic unit	Dark toned, smooth, crater-retaining unit with morphology similar to other exposures of volcanic plains on the surface of Mars [e.g., *Hiesinger and Head,* 2004; *Tanaka et al.,* 2014].	Coherent volcanic unit that bears similarities to Hesperian ridged plains and nearby Syrtis Major lava flows [*Hiesinger and Head,* 2004].	
PCu	Pitted capping unit	Relatively thin unit that caps older material. Has an eroded texture, a large amount of superposed craters, and pits of various sizes that often contain	Degraded, thin volcanic or impact melt unit that caps much of the altered basement, especially in the western fan watershed. May be	14

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Table 3. (continued)

Unit Map Symbol	Unit Name	Unit Description	Unit Interpretation	Relevant Figure(s)
		surficial debris cover (Ac). Unit forms mesas and appears relatively coherent.	related to older volcanic flows from nearby Syrtis Major [*Hiesinger and Head,* 2004].	
PCul	*Light-toned pitted capping unit*	*Has the same morphology as the pitted capping unit (PCu) but is lighter toned in CTX images and has no other morphologic distinctions.*	*Light toned and likely better exposed version of the pitted capping unit (PCu).*	14
MT	Mottled terrain	Light toned unit that appears heavily degraded and exhumed, and has a mottled texture. Displays an abundance of circular features that are likely to be degraded impact craters. Displays banding (not clear layering) in certain locations and appears relatively thin and remains in discrete patches throughout the watershed.	Part of a larger, regionally identified olivine- and carbonate-bearing unit [*Ehlmann et al.,* 2008b, 2009; *Mustard et al.,* 2009]. This is the only major unit observed both interior and exterior to the Jezero basin.	8–10 and 15
MTd	*Dusty mottled terrain*	*Has the same morphology as the mottled terrain unit (MT) but is more dust covered.*	*Dust-covered mottled terrain unit (MT).*	-
MTe	*Eroded mottled terrain*	*Has the same morphology as the mottled terrain unit (MT) but is more eroded.*	*Mottled terrain unit (MT) that is more heavily eroded than is typical.*	15
MTl	*Lineated mottled terrain*	*Has the same morphology as the mottled terrain unit (MT) but also has a lineated texture.*	*Mottled terrain unit (MT) that has a lineated texture that appears to be an erosional remnant.*	-
Bal	Altered basement	Light toned terrain that appears heavily eroded and exhumed, and has extremely variable topography on short length scales. Often forms small, light toned knobs or mesas.	Fe/Mg-smectite-rich, altered Noachian basement material that is heavily eroded and exhumed.	13
Balr	Ridged, altered basement	Has a similar morphology as the altered basement unit (Bal) but is also crosscut by short, quasi-linear ridges that appear more resistant to erosion than the surrounding terrain.	Altered basement unit (Bal) that is also crosscut by high-standing ridges, similar to those observed by *Head and Mustard* [2006] and *Saper and Mustard* [2013].	13, 14c, 15e, 15f, and 18a
Bdm	Dusty, massive basement	Light toned unit that appears massive (no obvious structure), dust covered, and is typically located in topographically high mounds or knobs.	Massive Noachian basement material that mostly appears dust covered, although in certain locations shows spectral signatures consistent with Fe/Mg-smectite, indicating it is at least partially altered.	12, 13c, and 13d

[a]The italics indicate subunits of the main geomorphic units.

4. Results

We mapped 28 distinct geomorphic units and subunits within the Jezero crater paleolake watershed (Figure 4a and Table 3) and basin (Figure 4c and Table 4). These units show a mineralogic and morphologic diversity that has been previously noted for the greater Nili Fossae region [e.g., *Mangold et al.,* 2007; *Mustard et al.,* 2007, 2009; *Ehlmann et al.,* 2009; *Ivanov et al.,* 2012]. We have also determined the approximate stratigraphy of these units (Figure 5). For the purpose of assessing the source-to-sink mineralogy of the Jezero crater paleolake system, we are only interested in those units within the basin and watershed that are stratigraphically below, or equivalent to, the fluvial features that drain into the paleolake. Within the basin, these represent preexisting units and lacustrine material with a sedimentary origin. Within the watershed, these represent potential source units for sediment supplied to the paleolake basin.

4.1. Jezero Basin Units

4.1.1. Jezero Crater Rim and Wall

Within the basin, the lowest stratigraphic unit is the Jezero crater rim and wall material (Crw; Figures 4c and 5b and Table 3) defined based on topography as well as the generally massive nature of the exposed portions (Figure 6). The Jezero crater rim is well exposed around approximately two thirds of the crater perimeter;

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Table 4. Geomorphic Units Mapped Only Within the Jezero Crater Paleolake Basin[a]

Unit Map Symbol	Unit Name	Unit Description	Unit Interpretation	Relevant Figure(s)
VF	Volcanic floor unit	Smooth, moderate toned, crater-retaining unit on the floor of Jezero crater. Unit appears relatively thin (<10 m along margins), is emplaced upon the light-toned floor unit (LTF), and embays the two fan deposits (Fn and Fw).	Volcanic unit that resurfaced the crater floor at ~3.45 [*Goudge et al.,* 2012b].	7 and 9–11
Fn	Northern fan deposit	Light toned, heavily eroded fan deposit at the mouth of the northern inlet valley in the Jezero basin. Deposit shows layering, although not at as fine scale as in the western fan (Fw).	Fan deposit formed at the mouth of the northern inlet valley.	9, 10, and 11e
Fw	Western fan deposit	Light toned, partially eroded and exhumed fan deposit at the mouth of the western inlet valley in the Jezero basin. Deposit shows fine-scale layering, as well as detailed sedimentary structures, such as preserved meanders and cross-bedding [*Fassett and Head,* 2005; *Schon et al.,* 2012].	Fan deposit formed at the mouth of the western inlet valley.	7, 9, and 11d
LTF	Light-toned floor unit	Light toned unit directly underlying the two fan deposits (Fn and Fw) and the volcanic floor unit (VF). Unit is commonly covered by aeolian dunes and appears pervasively fractured below the dunes.	Basin fill. Potentially mottled terrain (MT; Table 3) underlying the fan deposits (Fn and Fw) and volcanic floor unit (VF) and emplaced prior to lacustrine activity.	7, 9, 10, 11a, and 11b
LTFd	*Dusty light-toned floor unit*	*Similar morphology as the light-toned floor unit (LTF) but with an increased amount of dust cover.*	*Dust covered light-toned floor unit (LTF).*	-

[a]The italics indicate subunits of the main geomorphic units.

however, this unit is not identified in the north and northeast portions of the crater (Figure 4c). Generally, this unit appears spectrally bland (likely to be due to observable dust cover), although in small exposures we observe spectra with clear absorptions at ~1.41, 1.92, and 2.30–2.31 μm (Figure 6), which we interpret to be indicative of Fe/Mg-smectite (Figure 3).

4.1.2. Light-Toned Floor Unit and Mottled Terrain Unit

Stratigraphically above the Jezero crater rim and wall material are two units that occupy an equivalent stratigraphic position: the light-toned floor unit (LTF) and the mottled terrain unit (MT; Figures 4c and 5b and Tables 3 and 4). The light-toned floor unit is light toned, is often covered by aeolian dunes, and appears pervasively fractured when exposed below the dunes (Figure 7). This unit extends across much of the Jezero crater floor and is exposed in erosional windows below the stratigraphically higher volcanic floor unit (Figures 4c, 7a, and 7c).

The spectral signature of the light-toned floor unit is difficult to assess due to the presence of superposed dunes. The dunes have a distinct spectral signature with a strong, broad absorption centered near ~1 μm (Figure 7d) that is consistent with olivine (Figure 3). While the olivine-rich spectral signature of the dunes dominates most spectra from the light-toned floor unit, in select locations where there is minimal dune cover, spectra from this unit show additional narrow, vibrational absorptions at ~1.92 μm and paired absorptions at ~2.30 and 2.51 μm (Figure 7d). The paired absorptions at ~2.30 and 2.51 μm are consistent with the identification of carbonate, with the precise position of these absorptions best matched by a Mg-rich carbonate such as magnesite [*Hunt and Salisbury,* 1971; *Gaffey,* 1987] (Figure 3). We interpret the spectrum of the light-toned floor unit to indicate it is composed of a mixture of olivine and Mg-rich carbonate. The H_2O absorption at ~1.92 μm indicates that either the Mg-rich carbonate is hydrated [e.g., *Calvin et al.,* 1994], or that this unit contains an additional hydrated phase, such as a smectite mineral. The dunes that cover much of this unit are likely to be the erosional products of the light-toned floor unit, and so it is interesting to note that only olivine, and not carbonate, physically weathers out in grains that are conducive to the formation of dunes, consistent with the observations of *Mustard et al.* [2009].

Around the edges of the Jezero crater floor, the light-toned floor unit grades into the mottled terrain unit (MT; Figure 4c and Table 3). There is no clear stratigraphic relationship between these two units, so we interpret

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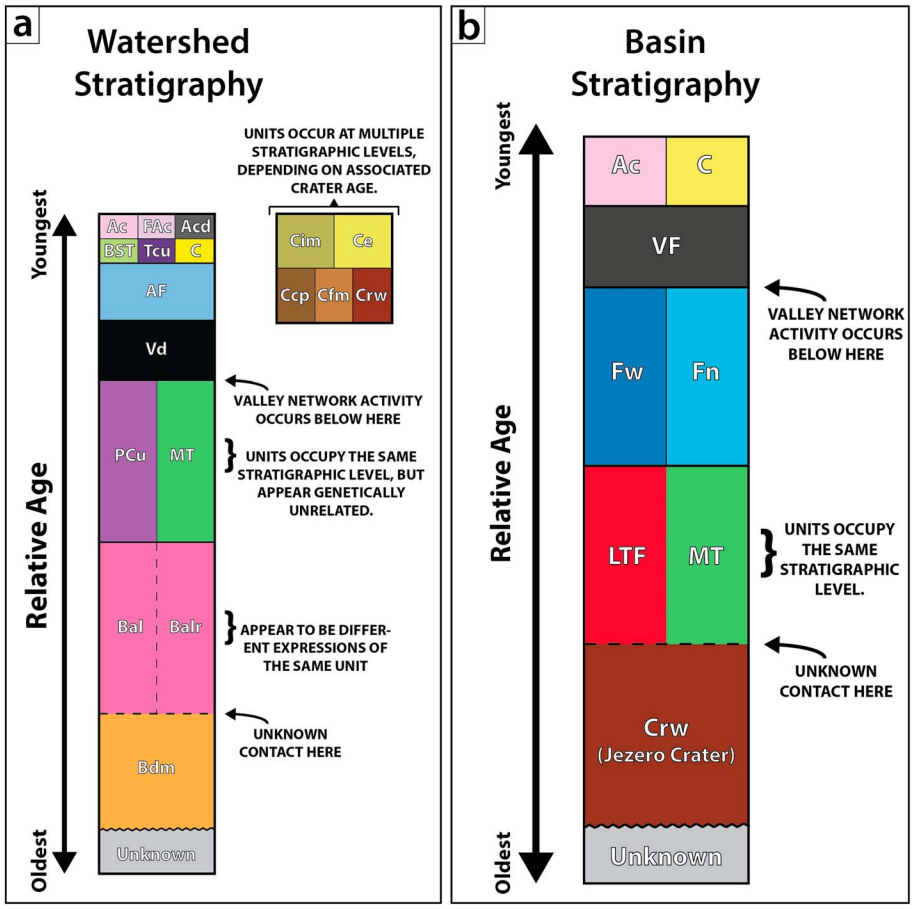

Figure 5. Generalized stratigraphic column for the geomorphic units within the Jezero crater (a) watershed and (b) paleolake basin. Units are listed in approximate stratigraphic order from youngest at the top to oldest at the bottom, except where noted. Also indicated is the stratigraphic level below which valley network activity occurred. Units are listed by their map symbol (see Tables 3 and 4).

them as stratigraphically equivalent (Figure 5b). The mottled terrain unit is the only major geomorphic unit identified both interior and exterior to the Jezero basin (Figure 4). The mottled terrain unit extends across the topographic boundary that defines the Jezero crater rim in the north and northeast (Figure 4c). In this region, there are no obvious exposures of the Jezero crater rim and wall material; however, the topographic signature of the rim and wall, while subdued, is still evident (Figure 1c).

The mottled terrain unit is light toned and characterized by a mottled texture and a heavily degraded and exhumed appearance with an abundance of circular features likely to be degraded impact craters (Figure 8). This unit is exposed with slightly variable morphology throughout the basin and watershed (Table 3), which we have classified into three subunits: mottled terrain with dust cover (MTd), eroded mottled terrain (MTe), and mottled terrain with lineated texture (MTl).

Where reasonably well exposed, the spectral signature of the mottled terrain unit shows a broad absorption centered near ~1 μm, as well as narrower, vibrational absorptions centered near ~1.92, 2.30–2.31, and 2.51 μm (Figure 8c). This spectral signature is very similar to that of the light-toned floor unit (Figure 7d) and is interpreted as representative of a mixture of olivine and Mg-rich carbonate, potentially with an additional hydrated phase (Figure 3).

4.1.3. Jezero Fan Deposits

Above the light-toned floor and mottled terrain units in the stratigraphic column are the western (Fw) and northern (Fn) sedimentary fan deposits (Figures 4c, 9, and 10 and Table 4). Both of these fan deposits are emplaced directly onto the light-toned floor unit (Figures 9e and 10e) and have previously been interpreted as delta deposits formed when Jezero crater hosted a paleolake [*Fassett and Head,* 2005; *Ehlmann et al.,*

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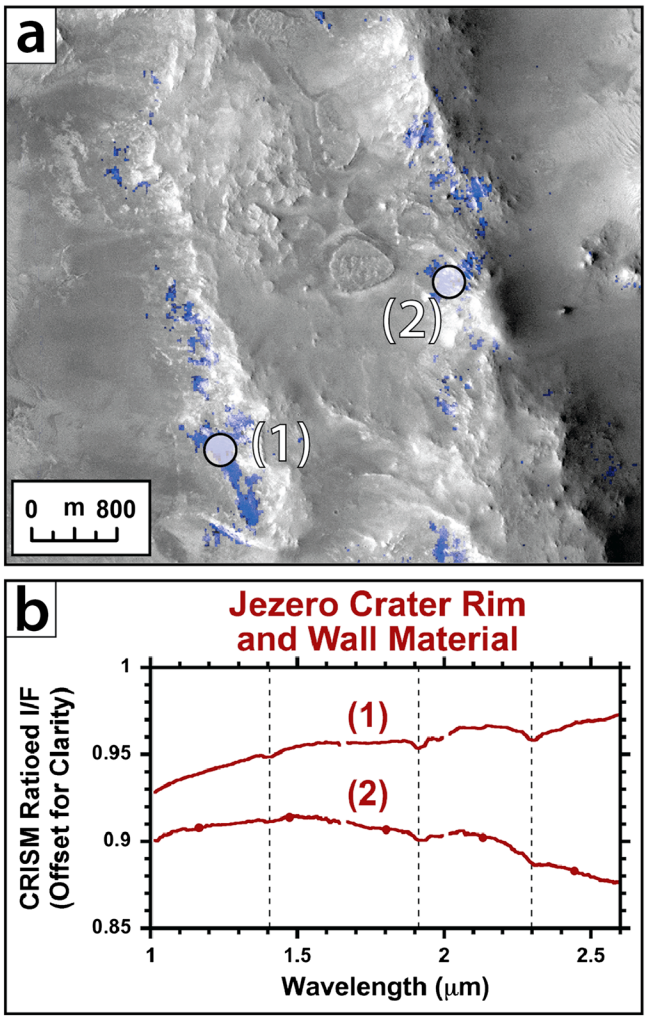

Figure 6. Jezero crater rim and wall material unit showing isolated locations with spectral signatures consistent with Fe/Mg-smectite. (a) CRISM-derived D2300 parameter map in blue [*Pelkey et al.,* 2007] from image FRT00005850 overlain on CTX image P04_002664_1988. Blue areas indicate regions consistent with Fe/Mg-smectite. North is up. (b) CRISM ratioed spectra of Fe/Mg-smectite exposures in the Jezero crater rim and will material unit. Numerator spectra are extracted from regions of interest in CRISM image FRT00005850 that are centered on labeled points shown in Figure 6a. Dashed lines are located at ~1.41, 1.92, and 2.3 μm.

2008a; *Schon et al.,* 2012]. Both fan deposits are heavily eroded (Figures 9 and 10), although the fate of the material eroded from the fans is not immediately apparent. There is a large area within the southeast portion of Jezero crater that contains abundant aeolian dunes (Figure 4c), and these dunes may be composed of sediment eroded from the fans; however, the dunes do not have the same spectral characteristics as the fans, and so a direct link is difficult to draw. It is also possible that the eroded fan material was transported out of the basin entirely via aeolian activity.

The western sedimentary fan deposit is the better preserved of the two deposits (Figure 9), showing inverted channels, well exposed sedimentary layering (Figures 9e and 9g), remnant meander loops, and evidence for fluvial cross-bedding [*Fassett and Head,* 2005; *Schon et al.,* 2012]. The western fan deposit ends in a well-defined, erosional scarp, suggesting the deposit once extended further into the basin center [*Schon et al.,* 2012]. Indeed, ~3 km from the western fan erosional front are a pair of small, light-toned, layered knobs (Figures 9c and 9f) interpreted to be remnants of the original fan deposit [*Schon et al.,* 2012].

In contrast, the northern fan deposit is more heavily eroded (Figure 10). The fan deposit does not extend completely back to the northern inlet valley mouth (Figures 4c and 10) but rather is exposed in a large outcrop, along with a few additional isolated patches, further toward the basin center. While these exposures do not show the same fluvial sedimentary structures as the western fan deposit, likely due to the higher degree of erosion, there is clear layering within exposures of this unit (Figure 10f), consistent with its interpreted origin as a sedimentary deposit.

The spectral properties of the western fan deposit are characterized by prominent vibrational absorptions at ~1.41, 1.91, and 2.31 μm (Figure 9d), interpreted to be representative of Fe/Mg-smectite (Figure 3); however, many of these spectra also have a distinct spectral shoulder (i.e., an abrupt change in spectral slope) or a very subtle absorption feature at ~2.51 μm, which we interpret to be caused by minor amounts of Mg-rich carbonate. The subtle spectral signature of Mg-rich carbonate in the dominantly Fe/Mg-smectite spectra may also be due to subpixel mixing, as there are small (tens of meters across) isolated regions on the western fan deposit that have spectra with a broad absorption at ~1 μm, as well as narrower absorptions at ~1.92, 2.30, and 2.51 μm (Figure 9d), interpreted to be caused by a mixture of Fe/Mg-smectite with olivine and Mg-rich carbonate (Figure 3). The spectral properties of the northern fan deposit are comparatively homogenous, characterized by a broad absorption centered near ~1 μm, and absorptions

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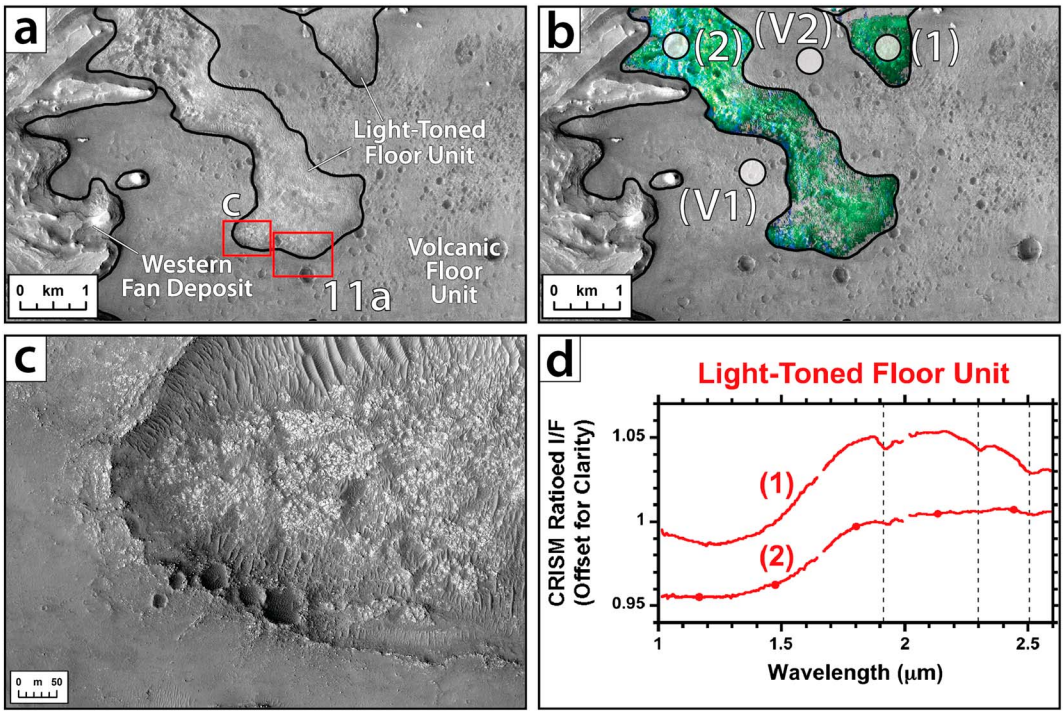

Figure 7. The light-toned floor unit. North is up in all images. (a) Geomorphic example of the light-toned floor unit. Note that the unit is light toned and has a high degree of cover by aeolian dunes. Also, note the smooth, crater-retaining appearance of the volcanic floor unit that is stratigraphically above the light-toned floor unit. Location of blowups (Figures 7c and 11a) are indicated by labeled red boxes. Unit contacts are indicated by thick black lines. Background is a portion of CTX image P04_002664_1988. (b) CRISM-derived parameter map (R = BD1900R, G = OLINDEX2, and B = D2300) [*Pelkey et al.,* 2007; *Ehlmann et al.,* 2009; *Salvatore et al.,* 2010] for the light-toned floor unit from image FRT000047A3 overlain on CTX image P04_002664_1988. Green areas indicate regions consistent with olivine, and cyan and white areas indicate regions consistent with a mixture of olivine and Mg-rich carbonate. Unit contacts are indicated by thick black lines. (c) Detailed example of the light-toned floor unit morphology. Note the pervasively fractured appearance of the unit underlying the darker, aeolian dunes. Portion of HiRISE image PSP_002743_1985. (d) CRISM ratioed spectra from the light-toned floor unit showing spectral signatures consistent with olivine (bottom spectrum) and a mixture of olivine and Mg-rich carbonate (top spectrum). Numerator spectra are extracted from regions of interest in CRISM image FRT000047A3 that are centered on labeled points shown in Figure 7b. Dashed lines are located at ~1.92, 2.3, and 2.51 μm.

at~1.92, 2.30–2.31, and 2.51 μm (Figure 10d), consistent with a mixture of olivine and either hydrous Mg-rich carbonate or anhydrous Mg-rich carbonate plus an additional hydrated phase, such as Fe/Mg-smectite (Figure 3).

4.1.4. Volcanic Floor Unit

The upper most major stratigraphic unit within the Jezero crater basin is the volcanic floor unit (VF), which lies stratigraphically above both of the fan deposits and the period of fluvial activity that fed the paleolake basin (Figures 4c and 5b and Table 4). The volcanic floor unit was deposited on top of the light-toned floor unit (Figures 7 and 11), and embays the western and northern fan deposits (Figures 11d and 11e) [*Goudge et al.,* 2012b; *Schon et al.,* 2012]. This unit is dark toned, relatively smooth, retains craters well (Figures 7 and 11), and appears to be <10 m thick, at least along its margins (Figure 11b). The CRISM-derived spectral signature of this unit shows two broad absorptions centered near ~1 and 2 μm (Figure 11c), which we interpret to be indicative of a mixture of olivine and pyroxene (Figure 3) in a basaltic material, consistent with previous work [*Goudge et al.,* 2012b]. There is no clear vent or fissure from which this volcanic floor unit might have been sourced; however, lack of a clear volcanic source is common for volcanically infilled crater on Mars, including volcanically resurfaced open-basin lakes [*Goudge et al.,* 2012b].

4.2. Jezero Watershed Units

Within the watershed of the Jezero crater paleolake system, there are four major geomorphic units that are important for assessing source-to-sink mineralogy (Figures 4a and 5a).

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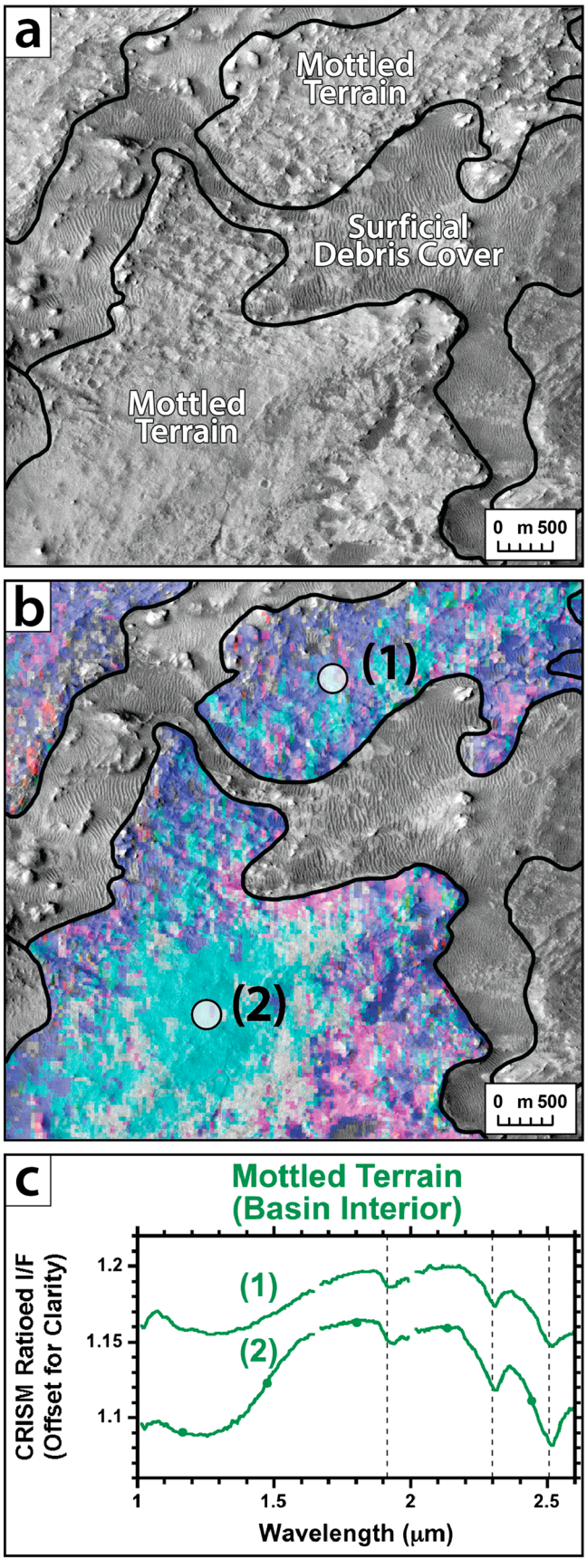

4.2.1. Dusty, Massive Basement Unit

A dusty, massive basement unit (Bdm; Figures 4a and 5a and Table 3) is stratigraphically lowest in the watershed and shows no discernable structure at CTX or HiRISE scale (Figure 12). Exposures of this unit are often in high-standing mounds or knobs and generally appear dust covered. The CRISM spectral properties of this unit are typically unremarkable; however, in small isolated regions in two overlapping CRISM images (FRT0000289E and HRL000095A2) in the northwest portion of the Jezero crater watershed, nearest the Nili Fossae trough (Figure 2), there are resolvable spectral absorption features. These spectra show vibrational absorptions at ~1.40, 1.92, and 2.31 μm (Figure 12b), which we interpret to be caused by Fe/Mg-smectite (Figure 3).

4.2.2. Altered Basement Units

Stratigraphically above the dusty, massive basement unit are the altered basement unit (Bal) and the ridged, altered basement unit (Balr), which are interpreted to be slightly different geomorphic expressions of the same fundamental unit (Figures 4a and 5a and Table 3). The altered basement unit is light toned and appears heavily eroded and exhumed, with rough, variable topography that often forms

Figure 8. The mottled terrain unit in the Jezero crater basin. North is up in all images. (a) Geomorphic example of the mottled terrain unit. Note that the unit is light toned and has an eroded, mottled surface texture. Unit contacts are indicated by thick black lines. Background is a portion of CTX image P04_002664_1988. (b) CRISM-derived parameter map (R = BD1900R, G = BD2500, and B = D2300) [*Pelkey et al.*, 2007; *Ehlmann et al.*, 2009] for the mottled terrain unit from image HRL000040FF overlain on CTX image P04_002664_1988. Cyan and white areas indicate regions consistent with Mg-rich carbonate. Unit contacts are indicated by thick black lines. (c) CRISM ratioed spectra from the mottled terrain unit showing spectral signatures consistent with a mixture of olivine and Mg-rich carbonate. Numerator spectra are extracted from regions of interest in CRISM image HRL000040FF that are centered on labeled points shown in Figure 8b. Dashed lines are located at ~1.92, 2.3, and 2.51 μm.

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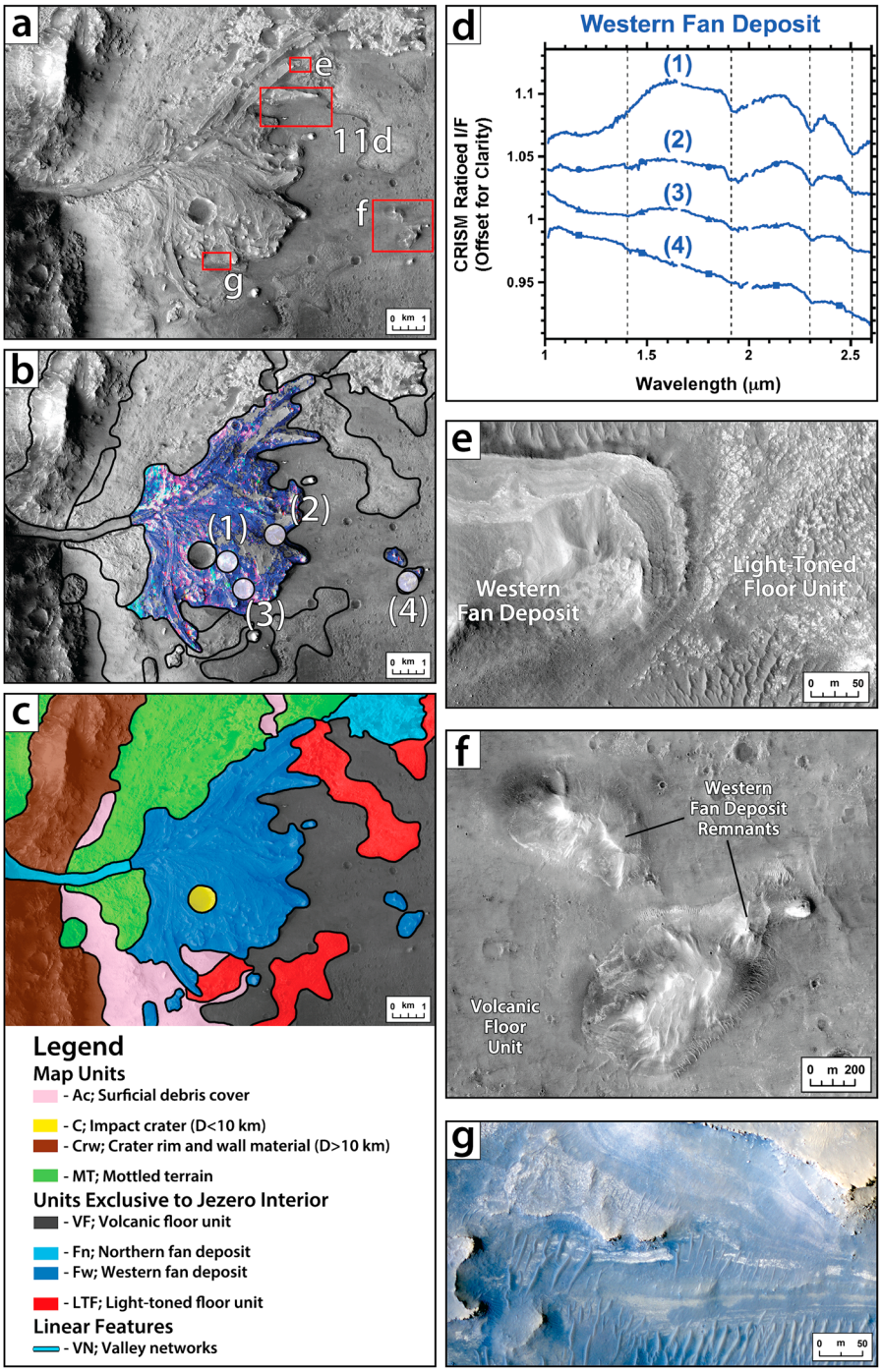

Figure 9. Jezero crater western fan deposit. North is up in all images. (a) Overview of the eroded but generally well preserved western fan deposit. Location of blowups (Figures 9e–9g and 11d) are indicated by labeled red boxes. Portion of CTX image P04_002664_1988. (b) CRISM-derived parameter map (R = BD1900R, G = BD2500, and B = D2300) [*Pelkey et al.*, 2007; *Ehlmann et al.*, 2009] for the western fan deposit from image HRL000040FF overlain on CTX image P04_002664_1988. Blue and pink areas indicate regions consistent with Fe/Mg-smectite, and cyan and white areas indicate regions consistent with Mg-rich carbonate. Unit contacts are indicated by thick black lines. (c) Geomorphic map of the Jezero crater western fan deposit, along with legend for units shown. Background is a portion of CTX image P04_002664_1988. (d) CRISM ratioed spectra from the western fan deposit showing spectral signatures consistent with a mixture of olivine and Mg-rich carbonate (top spectrum), and Fe/Mg-smectite (bottom three spectra). Spectra (2) and (3) are most representative of the western fan deposit, while the top spectrum (1) is representative of only discrete locations across the fan. The bottom spectrum (4) is from the western fan deposit remnants and displays a weak Fe/Mg-smectite signature. Numerator spectra are extracted from regions of interest in CRISM images HRL000040FF, spectra (1), (2), and (3), and FRT000047A3, spectrum (4), that are centered on labeled points shown in Figure 9b. Dashed lines are located at ~1.41, 1.92, 2.3, and 2.51 μm. (e) Detailed view of the western fan deposit superposing the light-toned floor unit. Portion of HiRISE image PSP_002387_1985. (f) Western fan deposit remnants, which display visible layering, and are embayed by the volcanic floor unit. Portion of HiRISE image ESP_023379_1985. (g) Example of layering with the western fan deposit. Portion of HiRISE IRB color image PSP_002387_1985.

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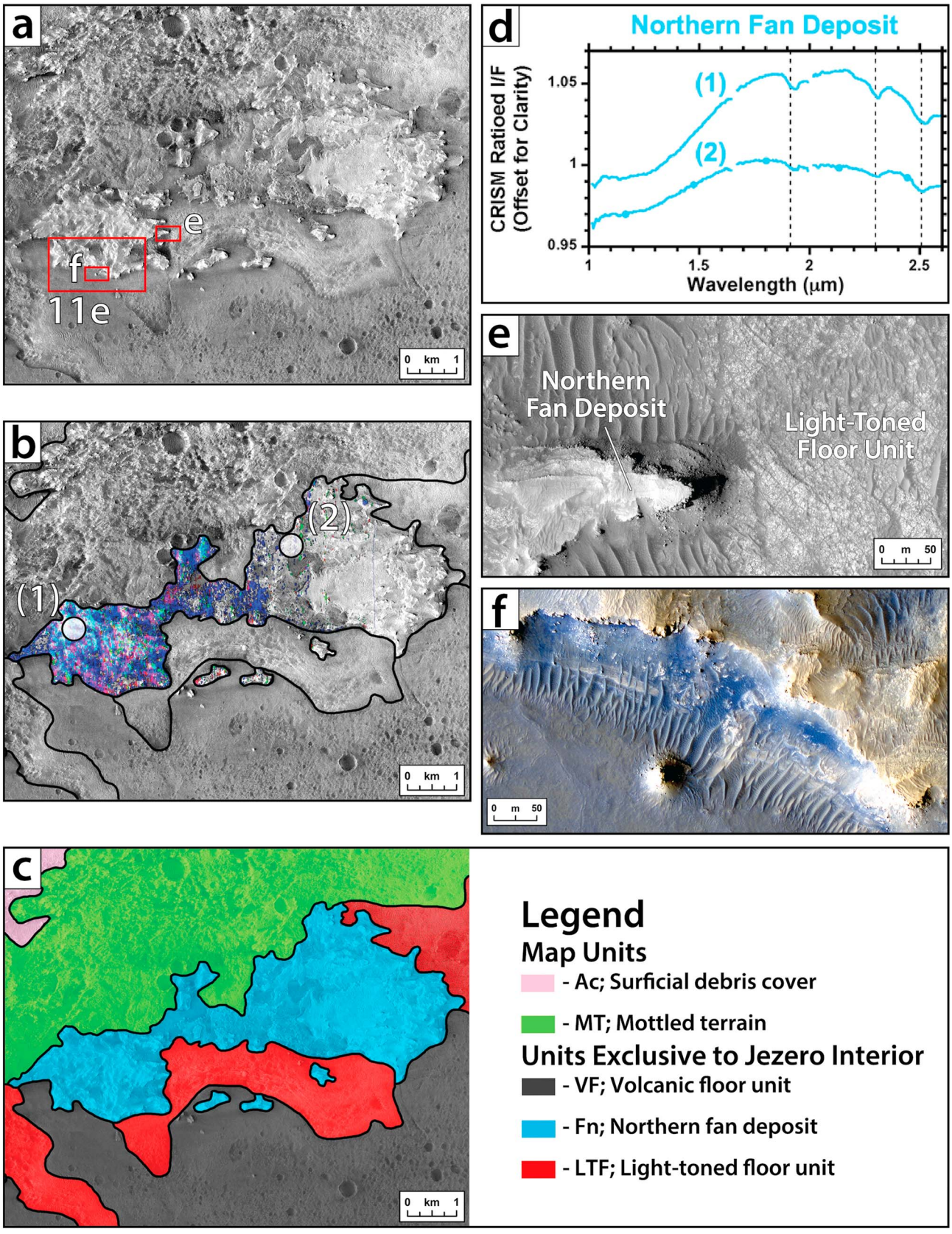

Figure 10

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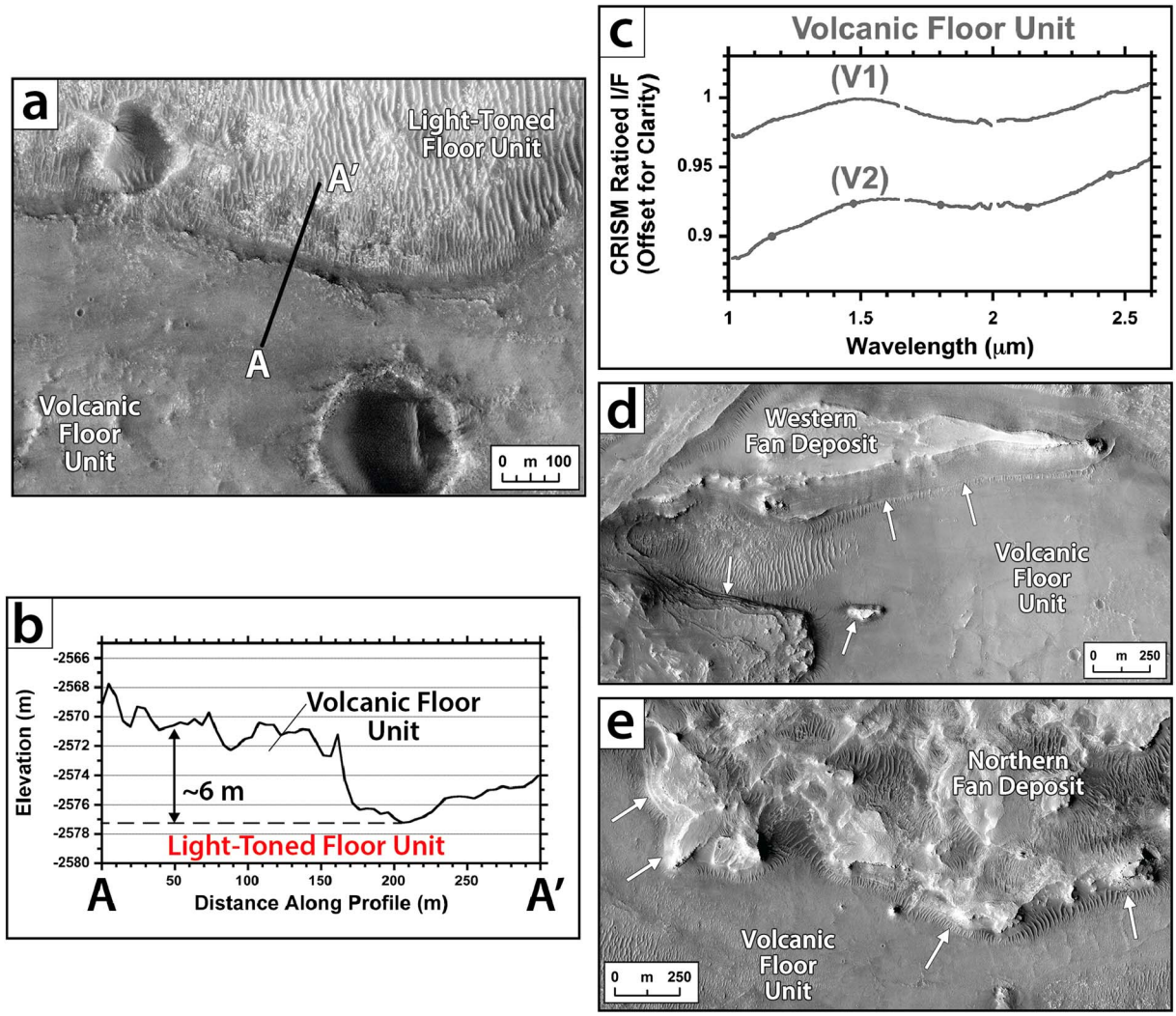

Figure 11. The volcanic floor unit. North is up in all images. (a) Example of the volcanic floor unit directly overlying the light-toned floor unit. Location of profile shown in Figure 11b is indicated by solid black line. Background is a portion of HiRISE image ESP_023524_1985. (b) Profile across the contact between the volcanic floor unit and the light-toned floor unit showing approximate thickness of the volcanic floor unit. Topographic data for profile are from a HiRISE stereo-derived DEM from image pair ESP_023379_1985 and ESP_023524_1985. (c) CRISM ratioed spectra from the volcanic floor unit showing spectral signatures consistent with a mixture of olivine and pyroxene. Numerator spectra are extracted from regions of interest in CRISM image HRL000040FF that are centered on labeled points shown in Figure 7b. (d) Example of the volcanic floor unit embaying the western fan deposit, indicated by white arrows. Mosaic of HiRISE images PSP_003798_1985 and PSP_002387_1985. (e) Example of the volcanic floor unit embaying the northern fan deposit, indicated by white arrows. Portion of HiRISE image PSP_002743_1985.

high-standing knobs or mesas (Figure 13). The ridged, altered basement unit has a very similar morphology, and it is also crosscut by quasi-linear ridges that are more resistant to erosion than the underlying terrain (Figure 13). These ridges are morphologically similar to ridges mapped by *Head and Mustard* [2006] and *Saper and Mustard* [2013] in the greater Nili Fossae region. In certain locations, both units show high concentrations

Figure 10. Jezero crater northern fan deposit. North is up in all images. (a) Overview of the heavily eroded, light toned northern fan deposit. Location of blowups (Figures 10e, 10f, and 11e) are indicated by labeled red boxes. Portion of CTX image P04_002664_1988. (b) CRISM-derived parameter map (R = BD1900R, G = BD2500, and B = D2300) [*Pelkey et al.,* 2007; *Ehlmann et al.,* 2009] for the northern fan deposit from image FRT000047A3 overlain on CTX image P04_002664_1988. Cyan, white, and green areas indicate regions consistent with Mg-rich carbonate. Unit contacts are indicated by thick black lines. (c) Geomorphic map of the Jezero crater northern fan deposit, along with legend for units shown. Background is a portion of CTX image P04_002664_1988. (d) CRISM ratioed spectra from the northern fan deposit showing spectral signatures consistent with a mixture of olivine and Mg-rich carbonate. Numerator spectra are extracted from regions of interest in CRISM image FRT000047A3 that are centered on labeled points shown in Figure 10b. Dashed lines are located at ~1.92, 2.3, and 2.51 μm. (e) Detailed view of the northern fan deposit superposing the light-toned floor unit. Portion of HiRISE image PSP_002743_1985. (f) Example of layering with the northern fan deposit. Portion of HiRISE IRB color image PSP_002743_1985.

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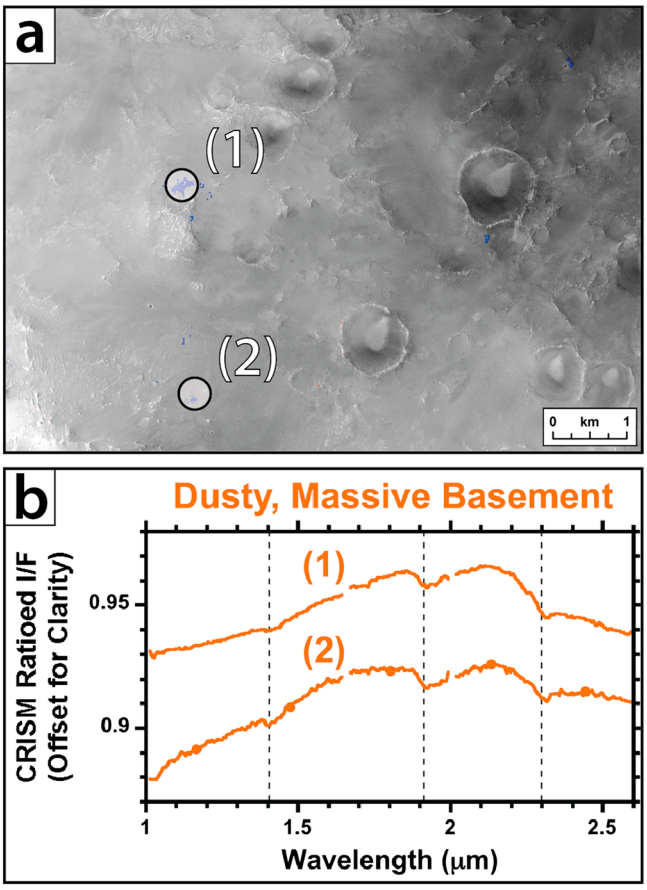

Figure 12. The dusty, massive basement unit. (a) Geomorphic example of the dusty, massive basement unit. Note the massive, smooth, dust-covered appearance of the unit. CRISM-derived D2300 parameter map in blue [*Pelkey et al.,* 2007] from image FRT0000289E overlain on CTX image P15_006844_2009. Blue areas indicate regions consistent with Fe/Mg-smectite. North is up. (b) CRISM ratioed spectra of the dusty, massive basement unit from isolated regions showing spectral signatures consistent with Fe/Mg-smectite. The majority of outcrops of the dusty, massive basement unit appear spectrally bland in CRISM data. Numerator spectra are extracted from regions of interest in CRISM image FRT0000289E that are centered on labeled points shown in Figure 12a. Dashed lines are located at ~1.41, 1.92, and 2.3 μm.

of brecciated material, including breccia blocks with distinct layering (Figure 13e). Spectra of the altered basement and ridged, altered basement units have distinct vibrational absorptions at ~1.41, 1.91, and 2.31 μm (Figure 13f), consistent with the presence of Fe/Mg-smectite (Figure 3). This spectral signature is fairly homogenous and observed in all locations where these units are well exposed.

4.2.3. Pitted Capping Unit

The pitted capping unit (PCu) and the mottled terrain unit (MT; Figures 4a and 5a and Table 3) were both emplaced directly upon the altered basement units. The pitted capping unit is relatively dark toned, has abundant superposed craters, appears generally coherent, and has an eroded texture that often forms pits of various sizes that are infilled by surficial debris (Figure 14). This unit is primarily observed in the western portions of the mapped region, and fluvial valleys that incise the pitted capping unit often erode into the underlying altered basement or ridged, altered basement units that are now exposed on the ancient valley floors (Figure 14c). CRISM spectra of this unit show two broad absorptions centered near ~1 and 2 μm (Figure 14d), consistent with electronic crystal field transitions from Fe^{2+} in olivine and pyroxene (Figure 3).

4.2.4. Mottled Terrain Unit

The mottled terrain unit (MT; Figures 4a and 5a and Table 3) is stratigraphically equivalent to the pitted capping unit. The mottled terrain unit is light toned with a mottled texture and is the only major unit observed both interior and exterior to Jezero crater. The morphology of the unit interior and exterior to the Jezero crater basin is very similar (Figures 8 and 15), although there are much larger exposures in the watershed than in the basin (Figure 4). In well exposed areas within the watershed, the mottled terrain unit shows a distinct banding (Figure 15f), which may be layering, although this determination cannot be made definitively with available HiRISE data. This banding is similar to banding observed in a regional olivine-rich unit identified by *Mustard et al.* [2009], who also note that this unit drapes the underlying alteration mineral-rich basement material, consistent with our observations. Exposures of the mottled terrain unit within the watershed appear to drape the underlying topography and also form distinct caps and mesas in surrounding exposures of the altered basement unit (Figures 4 and 15e).

The mottled terrain unit exposed in the watershed has a distinct and spatially homogenous spectral signature in CRISM data defined by a broad absorption centered near ~1 μm, with narrow, vibrational absorptions centered near ~1.92, 2.30–2.31, and 2.51 μm (Figure 15g). These properties are consistent with the spectral signature of the mottled terrain unit interior to Jezero crater (Figure 8c) and are interpreted to be indicative of a mixture of Mg-rich carbonate, olivine, and a potential additional hydrated mineral (Figure 3).

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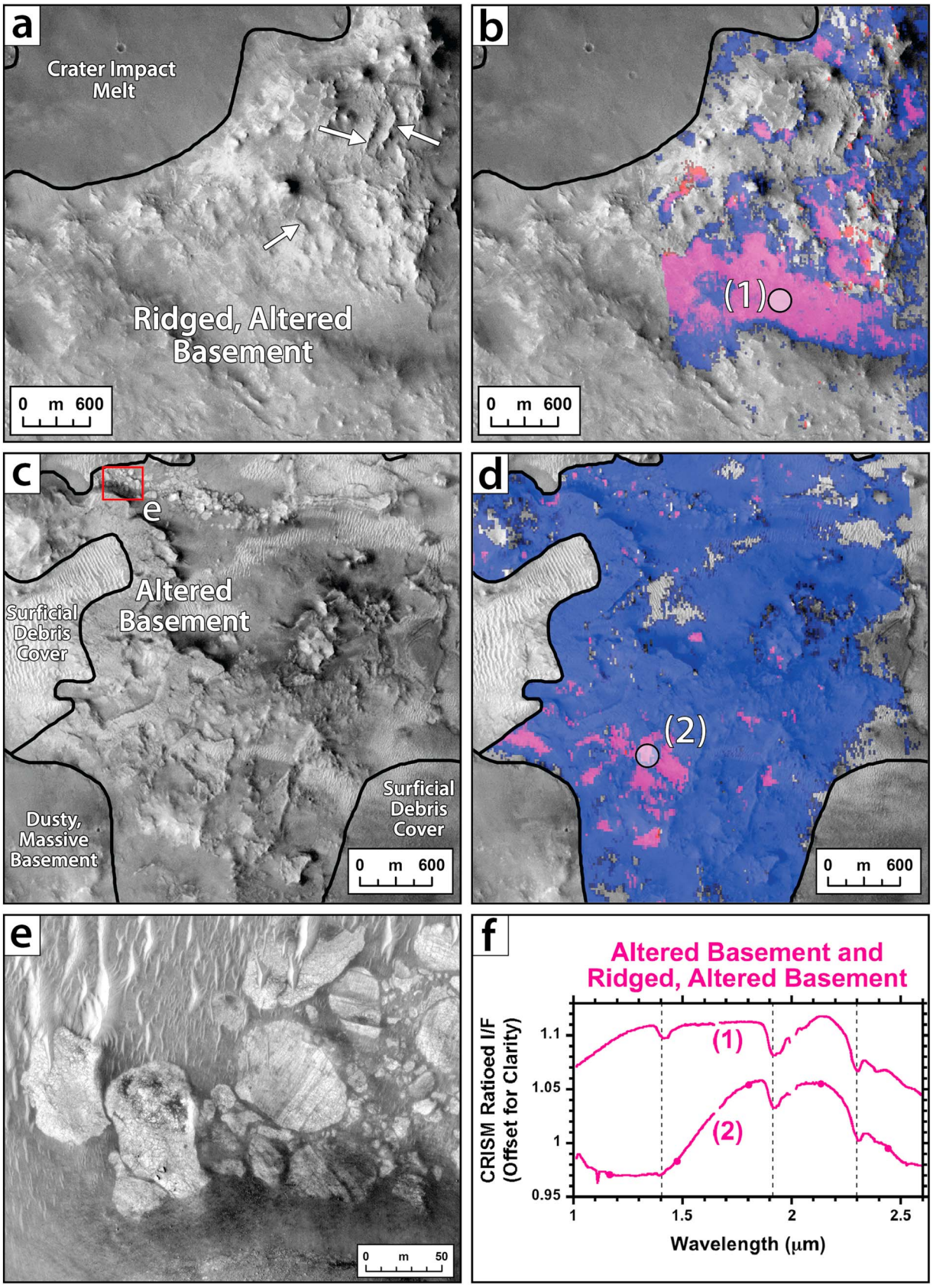

Figure 13

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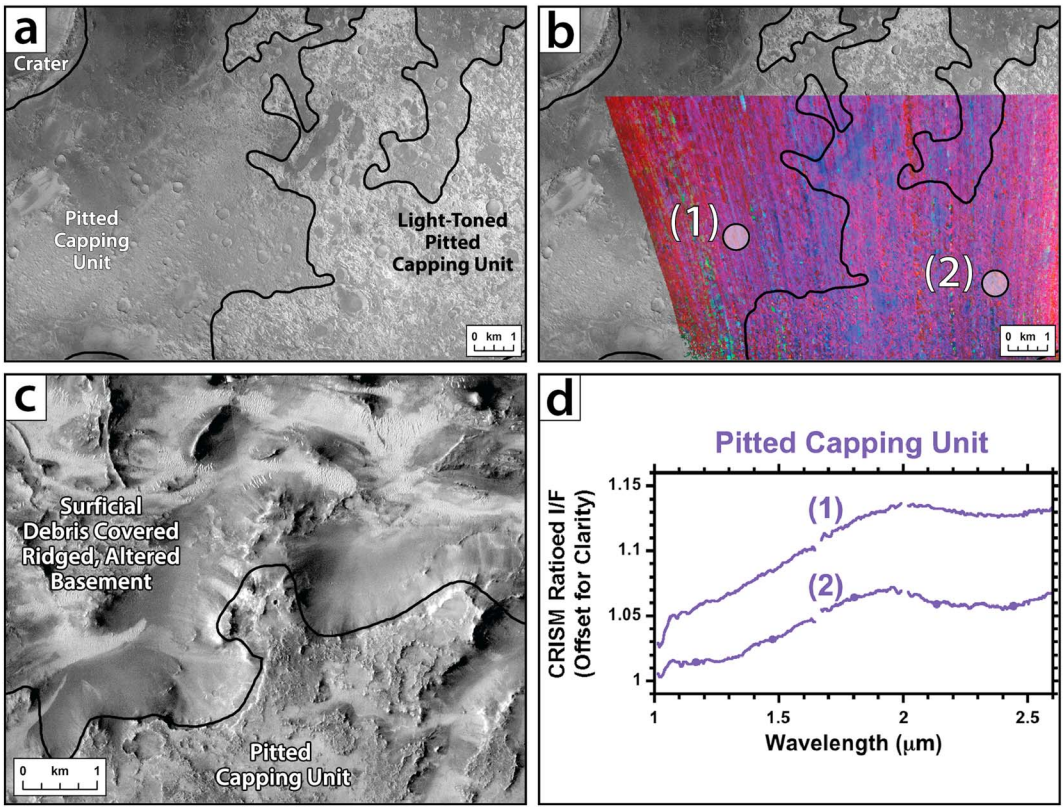

Figure 14. The pitted capping unit. North is up in all images. (a) Geomorphic example of the pitted capping unit and the light-toned pitted capping unit. Note the smooth, coherent nature of these units, with large amounts of superposed craters and surficial debris-filled pits. Unit contacts are indicated by thick black lines. Background is a mosaic of CTX images P08_004009_1982 and B03_010773_2000. (b) CRISM-derived parameter map (*R* = OLINDEX2, *G* = LCPINDEX, and *B* = HCPINDEX) [*Pelkey et al.,* 2007; *Salvatore et al.,* 2010] for the pitted capping unit from image FRT00013EBC overlain on a mosaic of CTX images P08_004009_1982 and B03_010773_2000. Bright colors (e.g., pinks and blues) indicate mafic-rich minerals. Unit contacts are indicated by thick black lines. (c) Example of the pitted capping unit overlying what is mapped as surficial debris cover in the floor of a valley network that fed the Jezero crater paleolake. Note that the surficial debris cover material is morphologically similar to the ridged, altered basement unit (see Figure 13), suggesting the valley network has eroded through the thin, pitted capping unit into the underlying ridged, altered basement unit. Unit contacts are indicated by thick black lines. Background is a portion of CTX image P19_008281_1982. (d) CRISM ratioed spectra from the pitted capping unit showing spectral signatures consistent with a mixture of olivine and pyroxene. Numerator spectra are extracted from regions of interest in CRISM image FRT00013EBC that are centered on labeled points shown in Figure 14b.

Figure 13. The altered basement and ridged, altered basement units. North is up in all images. (a) Geomorphic example of the ridged, altered basement unit. Note the light toned, heavily eroded terrain that forms small, high-standing knobs or mesas and the crosscutting ridges (indicated by white arrows). Unit contacts are indicated by thick black lines. Background is a portion of CTX image P05_002809_1975. (b) CRISM-derived parameter map (*R* = BD1900R, *G* = BD2200, and *B* = D2300) [*Pelkey et al.,* 2007; *Ehlmann et al.,* 2009] for the ridged, altered basement unit from image FRT000097E2 overlain on CTX image P05_002809_1975. Blue and pink areas indicate regions consistent with Fe/Mg-smectite. Unit contacts are indicated by thick black lines. (c) Geomorphic example of the altered basement unit. Note the light toned, heavily eroded terrain that forms small, high-standing knobs or mesas. Location of blowup (Figure 13e) is indicated by labeled red box. Unit contacts are indicated by thick black lines. Background is a portion of CTX image P15_006778_2002. (d) CRISM-derived parameter map (*R* = BD1900R, *G* = BD2200, and *B* = D2300) [*Pelkey et al.,* 2007; *Ehlmann et al.,* 2009] for the altered basement unit from image FRT000097E2 overlain on CTX image P15_006778_2002. Blue and pink areas indicate regions consistent with Fe/Mg-smectite. Unit contacts are indicated by thick black lines. (e) Megabreccia within the altered basement unit. Note the abundant breccia blocks of approximately tens of meters in scale, many of which display evidence of layering. Portion of HiRISE image PSP_006778_1995. (f) CRISM ratioed spectra from the ridged, altered basement unit, top spectrum (1), and the altered basement unit, bottom spectrum (2), showing spectral signatures consistent Fe/Mg-smectite. Numerator spectra are extracted from regions of interest in CRISM image FRT000097E2 that are centered on labeled points shown in Figures 13b and 13d. Dashed lines are located at ~1.41, 1.92, and 2.3 μm.

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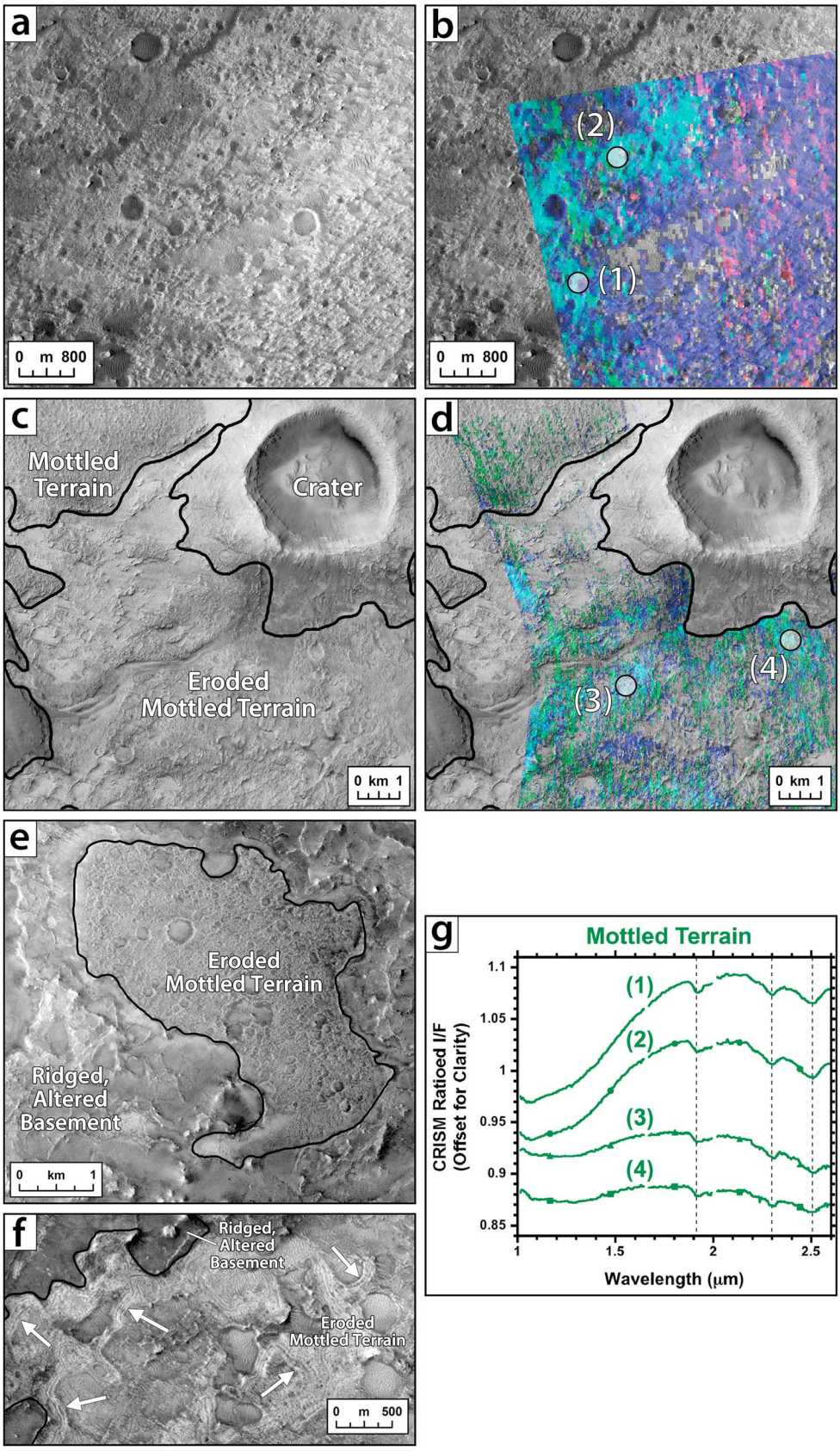

Figure 15

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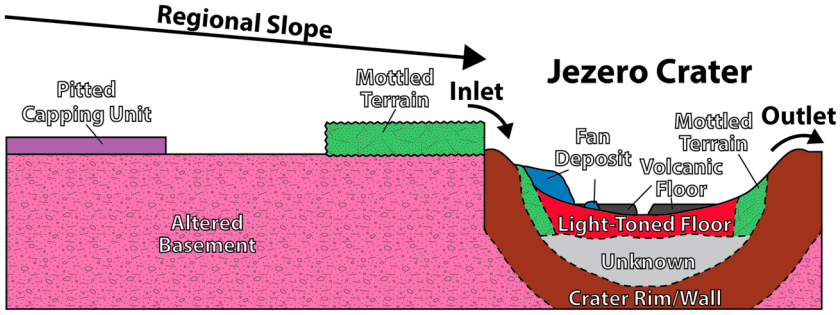

Figure 16. Generalized geologic model of the regional stratigraphy for the Jezero crater paleolake system. Transect is generally from the northwest of the Jezero crater paleolake watershed through the paleolake basin in the southeast.

5. Discussion

The results of our mapping lead to an interpreted stratigraphy of the Jezero crater basin (Figure 5b) and watershed (Figure 5a). This stratigraphy is used to develop a generalized model of the major geomorphic units present in the region (Figure 16). Using this geologic model, we can examine the source-to-sink mineralogy for the Jezero crater paleolake system, the regional geology, and its implications for the history of aqueous alteration in the Nili Fossae region of Mars.

5.1. Paleolake System Source-to-Sink Mineralogy

5.1.1. Jezero Crater Watershed Source Sediment

When assessing the source sediment for the Jezero crater paleolake system, we consider the four main units that are stratigraphically below the fluvial activity that fed the basin, defined as units that are either incised by the fluvial valleys or stratigraphically below such incised units. The four units are the pitted capping unit, the mottled terrain unit, the altered basement and ridged, altered basement units, and the dusty, massive basement unit (Figures 4, 5a, and 16 and Table 3).

We interpret the altered basement and ridged, altered basement units to be the primary sources for the Fe/Mg-smectite observed within the basin. Previous work on the mineralogy and stratigraphy of the Nili Fossae region has suggested that the widespread detections of Fe/Mg-smectite material represent the ancient, altered Noachian crust [*Mangold et al.,* 2007; *Mustard et al.,* 2007, 2009; *Ehlmann et al.,* 2009], an interpretation that we favor for our altered basement and ridged, altered basement units given the similarity in morphology and mineralogy. The observation of megabreccia within the altered basement and ridged, altered basement units (Figure 13e) was also noted by *Mustard et al.* [2009] for the more regional altered Noachian crust. *Mustard et al.* [2009] suggest that these breccia blocks are representative of both altered and unaltered crustal material derived from megaregolith excavated by the Isidis basin forming event, as well as brecciated Isidis ejecta, a hypothesis we favor here.

Figure 15. The mottled terrain unit in the Jezero crater watershed. North is up in all images. (a) Geomorphic example of the mottled terrain unit. Note that the unit is light toned and has an eroded, mottled surface texture. Portion of CTX image P04_002664_1988. (b) CRISM-derived parameter map (*R* = BD1900R, *G* = BD2500, and *B* = D2300) [*Pelkey et al.,* 2007; *Ehlmann et al.,* 2009] for the mottled terrain unit from image HRL000040FF overlain on CTX image P04_002664_1988. Cyan, white, and green areas indicate regions consistent with Mg-rich carbonate. (c) Geomorphic example of the mottled terrain unit and the eroded mottled terrain unit. Note that the unit is light toned and has an eroded, mottled surface texture, which is especially pronounced for the eroded mottled terrain unit. Unit contacts are indicated by thick black lines. Background is a portion of CTX image P17_007780_1987. (d) CRISM-derived parameter map (*R* = BD1900R, *G* = BD2500, and *B* = D2300) [*Pelkey et al.,* 2007; *Ehlmann et al.,* 2009] for the mottled terrain unit from image FRT0001182A overlain on CTX image P17_007780_1987. Cyan, white, and green areas indicate regions consistent with Mg-rich carbonate. Unit contacts are indicated by thick black lines. (e) Example of an isolated cap of the eroded mottled terrain unit overlying the ridged, altered basement unit. Unit contacts are indicated by thick black lines. Background is a portion of CTX image P15_006712_2010. (f) Example of banding (white arrows) in the mottled terrain unit. Unit contacts are indicated by thick black lines. Background is a portion of CTX image P21_009151_2014. (g) CRISM ratioed spectra from the mottled terrain unit showing spectral signatures consistent with a mixture of olivine and Mg-rich carbonate. Numerator spectra are extracted from regions of interest in CRISM images HRL000040FF, spectra (1) and (2), and FRT0001182A, spectra (3) and (4), that are centered on labeled points shown in Figures 15b and 15d. Dashed lines are located at ~1.92, 2.3, and 2.51 μm.

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The dusty, massive basement unit is interpreted to be either an unaltered component of the Noachian crust and/or a dust covered and poorly exposed portion of the altered Noachian crust. Given the major difference in morphology between the dusty, massive basement unit (Figure 12) and the altered basement and ridged, altered basement units (Figure 13), we interpret the dusty, massive basement unit as a distinct portion of the Noachian crust; however, it is difficult to state with certainty whether this portion of the Noachian crust is altered or unaltered. The dusty, massive basement unit appears spectrally bland in most locations with CRISM data, although in at least select, isolated locations this unit displays spectral signatures consistent with Fe/Mg-smectite (Figure 12b). Therefore, we suggest that the dusty, massive basement unit is likely to contribute either spectrally bland sediment to the Jezero crater paleolake basin or perhaps also some component of Fe/Mg-smectite.

The mottled terrain unit within the watershed (Figure 15) is the primary potential source for the Mg-rich carbonate and olivine mixture within the light-toned floor and fan deposit units within the Jezero basin. Identifications of Mg-rich carbonate in Nili Fossae, including in the mottled terrain unit, are consistently found associated with olivine [*Ehlmann et al.,* 2008b, 2009; *Mustard et al.,* 2009]. It has been suggested that the Mg-rich carbonate formed from the aqueous alteration of a regional olivine-rich unit that drapes the underlying topography [*Ehlmann et al.,* 2008b, 2009; *Mustard et al.,* 2009], a conclusion that is supported by our analysis of the Jezero crater watershed. The protolith olivine-rich unit has variably been interpreted as having been formed via exposure of an olivine-rich subsurface layer (e.g., an intrusion) [*Hoefen et al.,* 2003], olivine-rich basaltic flows [*Hamilton and Christensen,* 2005], or emplacement of an olivine-rich impact melt sheet associated with the formation of the Isidis impact basin [*Mustard et al.,* 2007, 2009]. *Ehlmann et al.* [2008b, 2009] suggest several mechanisms for the alteration of the olivine-rich protolith to Mg-rich carbonate, including hydrothermal alteration or serpentinization reactions either in the subsurface or at the surface, or surface alteration at ambient temperatures.

The pitted capping unit has a spectral signature characterized by a mixture of olivine and pyroxene (Figure 14d), and so is likely to have contributed sediment with this spectral character to the Jezero crater paleolake basin; however, it is important to note that a substantial length of the fluvial valleys that fed the Jezero crater paleolake system erode into, and through, the pitted capping unit (Figure 4), exposing the phyllosilicate-bearing altered basement units on the valley network floor (Figure 14c). Therefore, erosion of the pitted capping unit would supply mafic-rich source sediment to the Jezero crater paleolake basin, and incision through this unit by the valley networks that fed the basin would also supply Fe/Mg-smectite-rich sediment from erosion into the underlying altered basement unit. The origin of the pitted capping unit is not directly evident; however, given the basaltic spectral signature (Figure 14d) and the coherent nature of this unit (Figure 14), we interpret it as either impact melt from the Isidis basin forming event [e.g., *Mustard et al.,* 2007, 2009] or a volcanic unit, potentially related to the nearby Syrtis Major volcanic complex [*Hiesinger and Head,* 2004; *Tanaka et al.,* 2014].

5.1.2. Jezero Crater Fan Deposits

From our mapping and analysis of CRISM data, we conclude that the western fan deposit is spectrally dominated by Fe/Mg-bearing smectite with isolated exposures of Mg-rich carbonate and olivine (Figure 9d), while the northern fan deposit is spectrally dominated by a mixture of Mg-rich carbonate, olivine, and potentially an additional hydrated phase, such as a smectite mineral (Figure 10d). It is interesting to note that in addition to being composed of distinct mineral components, these two fan deposits have very different states of degradation (Figures 9 and 10). It is possible that the two fan deposits formed at different times, and that the difference in degradation states is related to the age of the deposits (i.e., the northern fan deposit formed before the western fan deposit); however, we find no evidence to support this hypothesis. Therefore, we conclude that both fan deposits formed during the same interval of fluvial activity, which ceased by ~3.8 Ga [*Fassett and Head,* 2008a].

Instead, we hypothesize that the difference in degradation states of the two fans is related to the different lithology of these two deposits. This implies that the olivine and Mg-rich carbonate-dominated northern fan deposit is less resistant to erosion than the Fe/Mg-smectite-dominated western fan deposit. A separate observation that may support this inference is the olivine-rich nature of the dunes derived from the olivine- and Mg-rich carbonate-bearing light-toned floor unit (Figure 7). This observation suggests that erosion of the light-toned floor unit yields only olivine particles large enough to form aeolian dunes, whereas Mg-rich carbonate particles eroded from this unit must be either present as, or eroded into, grains too small to form dunes. This would suggest that the Mg-rich carbonate phase in these deposits is generally poorly lithified and

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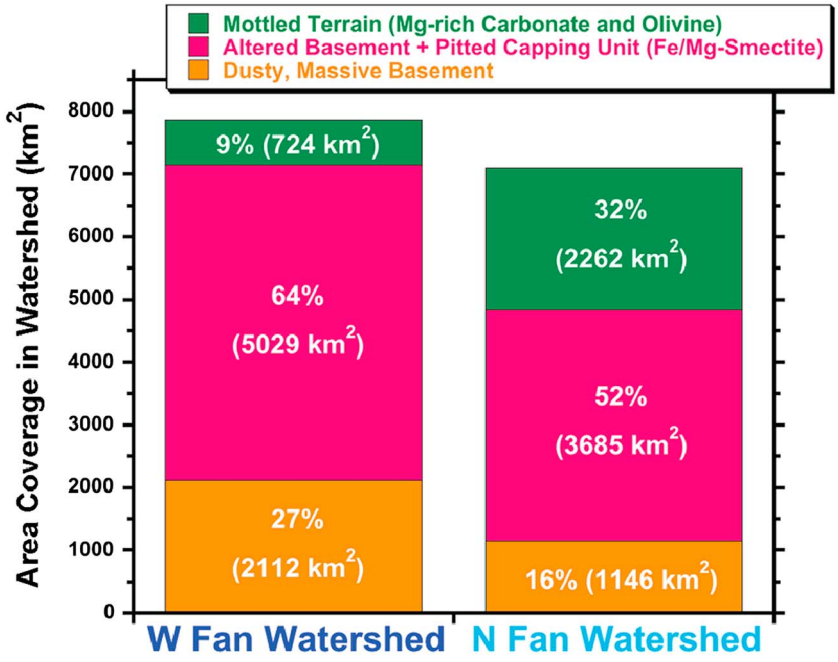

Figure 17. Histogram of the areal extent within the Jezero crater paleolake watershed of the major geomorphic units that are stratigraphically below the fluvial activity that fed the basin.

friable in nature, which may support the idea of a northern fan deposit that is less resistant to erosion than the Fe/Mg-smectite-dominated western fan deposit.

The difference in observed fan deposit mineralogy also provides important context when testing the hypothesis that the alteration phases observed within these sedimentary deposits are detrital in nature [*Ehlmann et al.,* 2008a, 2008b, 2009]. If the alteration phases observed within the two Jezero fan deposits are primarily transported, or detrital, in origin, then the difference in mineralogy between these two sedimentary fan deposits is likely to be due to a difference in the source sediment within their respective watersheds, i.e., their provenance. Here we test this hypothesis by considering the spectral signatures and extent of exposure (i.e., area of map coverage) of the major potential source sediment units within their watersheds (Figure 4). Given that the valleys mapped in the pitted capping unit commonly erode through this unit and expose the underlying altered basement unit on the valley floor (Figure 14c), we have grouped these geomorphic units together for area calculations.

The areal extent of exposure of the major geomorphic units within the two fan deposit watersheds was calculated from our geomorphic map (Figure 4) using an equal-area projection in the ArcMap GIS software (Figure 17). In the watershed of the western fan deposit, the dusty, massive basement unit occupies an area of ~2112 km^2 (~27% of the watershed excluding the area covered by geomorphic units stratigraphically above fluvial activity); the altered basement, ridged, altered basement, and pitted capping units occupy an area of ~5029 km^2 (~64%); and the mottled terrain unit occupies an area of ~724 km^2 (~9%; Figure 17). In the watershed of the northern fan deposit, the dusty, massive basement unit occupies an area of ~1146 km^2 (~16%); the altered basement, ridged, altered basement, and pitted capping units occupy an area of ~3685 km^2 (~52%); and the mottled terrain unit occupies an area of ~2262 km^2 (~32%; Figure 17).

The Mg-rich carbonate- and olivine-bearing mottled terrain unit is far more widespread (covers ~3 times the area) in the watershed of the northern fan deposit compared to the western fan deposit. The western fan deposit watershed is predominantly composed of the Fe/Mg-smectite-bearing altered basement unit, the ridged, altered basement unit, and the directly overlying pitted capping unit. Therefore, the observed difference in the spectral signature of the two fan deposits within the Jezero crater paleolake (Figures 9d and 10d) primarily reflects the mineralogy and spectral characteristics of the major geomorphic units within the different watersheds of these two deposits. This provides strong evidence that the spectrally dominant components of the two fan deposits, including the aqueous alteration phases, are primarily transported in

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nature, and that there has been little in situ alteration or mineral precipitation in this paleolake system, consistent with previous work at this site [*Ehlmann et al.,* 2008a, 2008b, 2009].

The occurrence of detrital phyllosilicates or clay minerals within the Jezero crater paleolake basin is to be expected given the phyllosilicate-rich nature of the exterior Nili Fossae terrain [e.g., *Mangold et al.,* 2007; *Mustard et al.,* 2007, 2009; *Ehlmann et al.,* 2009], as well as the prevalence of detrital clay minerals in terrestrial lacustrine sediment [e.g., *Moore,* 1961; *Thomas et al.,* 1972; *Jones and Bowser,* 1978]. Carbonate in terrestrial lake deposits, however, is more commonly precipitated from the water column as opposed to being transported into the basin from the surrounding watershed [e.g., *Brunskill,* 1969; *Jones and Bowser,* 1978; *Kelts and Hsu,* 1978; *Strong and Eadie,* 1978]. Nevertheless, detrital carbonate is often observed in terrestrial lake deposits in areas with large, regional exposures of carbonate-rich terrain (e.g., karst terrain) [e.g., *Jones and Bowser,* 1978; *Kelts and Hsu,* 1978; *Deevey et al.,* 1983; *Martin-Puertas et al.,* 2011], a situation that may be analogous to the widespread, regional, carbonate-bearing mottled terrain unit present in the Jezero crater paleolake watershed. Therefore, we interpret our source-to-sink characterization of the Jezero crater paleolake system to indicate that the mineralogy of the two fan deposits is primarily reflective of detrital sediment, and that there was no major aqueous alteration or mineral precipitation in situ or during sediment transport.

In terrestrial lacustrine environments, large-scale in situ formation of aqueous alteration and diagenetic minerals is most common in alkaline, saline lakes [e.g., *Singer and Stoffers,* 1980; *Jones and Weir,* 1983; *Jones*, 1986; *Deocampo et al.,* 2009], and so our observations suggest that the Jezero crater paleolake was neither highly alkaline nor highly saline. This conclusion is also consistent with the hydrologic environment of this open-basin paleolake [*Fassett and Head,* 2005, 2008b; *Schon et al.,* 2012]. Additionally, the presence of both Fe/Mg-smectite and carbonate within this paleolake basin suggests moderate to neutral pH conditions [*Ehlmann et al.,* 2008a, 2008b]. Taken together, this suggests that the Jezero crater paleolake was likely to have been a habitable environment during at least the time period of formation of the two fan deposits.

5.1.3. Carbonate-Bearing Materials in Jezero Crater

In addition to the two sedimentary fan deposits, Jezero crater contains two other units of potential lacustrine origin. These are the light-toned floor unit and the mottled terrain unit (Figures 4c, 5b, and 16), both of which are spectrally dominated by a mixture of Mg-rich carbonate and olivine (Figures 7d and 8c). These units are stratigraphically equivalent, and there is no clear contact between the two, but rather the light-toned floor unit grades into the mottled terrain unit. The light-toned floor unit is found primarily on the Jezero crater floor (Figure 4c), while the mottled terrain unit is found in a "ring" around the edges of the crater floor, as well as exterior to the basin (Figures 4a and 4c).

Both the light-toned floor and mottled terrain units are light toned with minimal observable structure (Figures 7 and 8), although the mottled terrain unit displays banding in select locations within the watershed (Figure 15f). Additionally, the mottled terrain unit has highly variable topography on short length scales (Figure 8), giving it a mottled texture, while the light-toned floor unit has more subdued topography, is pervasively fractured, and is commonly covered by aeolian dunes (Figure 7).

The mottled terrain unit extends across the northern Jezero crater rim with no change in its geomorphic expression (Figure 4). We thus conclude that the mottled terrain unit interior and exterior to Jezero crater has the same origin, i.e., the aqueous alteration of a previously existing olivine-rich unit [*Ehlmann et al.,* 2008b, 2009; *Mustard et al.,* 2009]. The origin of the light-toned floor unit is slightly less clear, and we suggest two primary hypotheses for the formation of this unit. First, it is possible that the light-toned floor unit is a different geomorphic expression of the mottled terrain unit. In this scenario, the light-toned floor unit would have been emplaced prior to the lacustrine activity within the Jezero crater paleolake.

Alternatively, it is possible that the light-toned floor unit represents sedimentary fill emplaced during the lacustrine activity within Jezero crater. In this scenario, the light-toned floor unit would consist of either detrital carbonate-rich sediment or carbonate precipitated from the water column within the Jezero crater paleolake; however, definitively establishing an in situ origin for the light-toned floor unit using only orbital data would prove extremely difficult. This scenario requires that the light-toned floor unit is stratigraphically above the mottled terrain unit, and that the contact between these two units has been obscured.

Given the overall similarity in morphology (Figures 7 and 8), mineralogy as inferred from CRISM data (Figures 7d and 8c), and lack of any clear contact or stratigraphic relationship between the light-toned floor

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and mottled terrain units, we favor the hypothesis that these two geomorphic units are different expressions of the same geologic unit. While it is possible that the light-toned floor unit is lacustrine in origin, none of the data we have examined support this conclusion.

This hypothesis implies that the basin fill within Jezero crater was deposited prior to the lacustrine activity within the basin, as opposed to being lacustrine in origin as assumed by *Schon et al.* [2012]. If this hypothesis is correct, then the maximum duration of lacustrine activity of ~10^6–10^7 years within Jezero crater estimated by *Schon et al.* [2012] is likely to be a substantial overestimate. However, detailed interpretations of meander and scroll bar migration from the western fan provide a minimum estimate of ~20–40 years for lacustrine activity in the Jezero crater paleolake [*Schon et al.,* 2012].

Any deep basin lacustrine sediment deposited in Jezero crater must have been emplaced upon the light-toned floor unit; however, as the volcanic floor unit is emplaced directly onto the light-toned floor unit (Figure 11), this implies that the majority of basinal lacustrine sediment in Jezero crater was eroded subsequent to the end of lacustrine activity within the basin but prior to the emplacement of the volcanic floor unit. *Fassett and Head* [2008a] estimate an age of cessation for the valley networks that fed Jezero crater of ~3.8 Ga, while *Goudge et al.* [2012b] estimate an age for emplacement of the volcanic floor unit of ~3.45 Ga. This gives ~300–400 Myr to erode any lacustrine material deposited in the basin interior, although with the error estimates on the ages from *Fassett and Head* [2008a] and *Goudge et al.* [2012b], this length of time could range from ~100 to 1100 Myr. Regardless of the absolute amount of time elapsed between the end of fluvial activity and the emplacement of the volcanic floor unit, based upon the heavily eroded nature of the two sedimentary fan deposits in the basin (Figures 9 and 10) [*Fassett and Head,* 2005; *Schon et al.,* 2012], it seems reasonable to suggest that any lacustrine sediment deposited in the basin interior, which would have been of a finer grain size than the two fan deposits [e.g., *Rich,* 1951; *Michael et al.,* 2014], was completely eroded and either transported out of the basin or remains in the crater as surficial cover/aeolian dunes.

5.2. Implications for Regional Aqueous Alteration

From our source-to-sink study of the mineralogy of the Jezero crater paleolake system, we have concluded that the alteration minerals observed within the two fan deposits are transported in origin, with no evidence for in situ alteration. This conclusion requires that the alteration mineral-bearing source units for detrital sediment transported to the Jezero basin must have been present in the watershed prior to erosion, transport, and deposition in the fan deposits. An implicit assumption in this conclusion is that the period of alteration that formed the Fe/Mg-smectite in the altered basement and ridged, altered basement units, and the alteration that formed the Mg-rich carbonate within the mottled terrain unit must have occurred prior to the period of fluvial erosion in the watershed—the alteration minerals could only be transported as detrital sediment if they had been formed in the watershed area.

Based on our mapping results and interpreted stratigraphy (Figures 4, 5, and 16), we argue that this is a valid assumption and is most consistent with our observations. The strongest argument for this hypothesis is the widespread identification of the pitted capping unit. This unit has an unaltered, basaltic composition based on its spectral signature (Figure 14d); however, we note that the pitted capping unit has only minimal CRISM coverage (images HRL000095A2, FRT0000289E, FRT00013EBC, and FRT00023370; Figure 2 and Table 2). Although this is the case, the unaltered, basaltic spectral signature of the pitted capping unit is consistent across all four images, and yet this unit is clearly incised by the valley networks that fed the Jezero crater paleolake (Figures 4 and 14c). If the alteration minerals observed in the Jezero crater basin and watershed formed either during or subsequent to the fluvial activity that fed this paleolake, it would be expected that the pitted capping unit should also have been altered to some degree; however, we have found no evidence for alteration of this unit anywhere within the Jezero crater watershed.

Furthermore, if the chemical weathering associated with the formation of the alteration minerals occurred during the period of fluvial activity that fed the Jezero crater paleolake, this should be reflected in the sedimentary fan deposits within the basin. For contemporaneous alteration and fluvial activity, the fan deposits would be expected to display unaltered material in more basal units deposited at the onset of fluvial activity and sediment transport, but prior to the buildup of alteration minerals in the watershed, transitioning to more altered material at higher (younger) stratigraphic levels of the deposits. However, none of our analyses indicate such a mineralogic transition, and instead it appears that the two fan deposits have a spectrally dominant

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component of alteration minerals (primarily Fe/Mg-smectite and Mg-rich carbonate for the western and northern fans, respectively) regardless of stratigraphic position within the depositional section.

Therefore, we conclude that the period of fluvial activity associated with the Jezero crater paleolake system, estimated to have ceased at ~3.8 Ga [*Fassett and Head,* 2008a], is not genetically related to the regional aqueous alteration in Nili Fossae [*Mangold et al.,* 2007; *Mustard et al.,* 2007, 2009; *Ehlmann et al.,* 2009]. This conclusion requires at least two periods of aqueous activity early in Martian history associated with the Jezero crater paleolake watershed: (1) a period of aqueous alteration to form the Fe/Mg-smectite and Mg-rich carbonate, and (2) a period of surface runoff, fluvial erosion, and lacustrine activity.

Two distinct periods of aqueous activity are consistent with the work of *Ehlmann et al.* [2009], as well as the geologic models for early Mars presented by *Ehlmann et al.* [2011] and *Fassett and Head* [2011]. As proposed by *Ehlmann et al.* [2011], the widespread presence of Fe/Mg-rich clay minerals suggests that the period of early aqueous alteration occurred via closed-system weathering in the subsurface, which would not require warm and wet surface conditions. In this model, the early, subsurface aqueous activity would have been responsible for the alteration that formed the Fe/Mg-smectite in the altered basement and ridged, altered basement units within the Jezero crater watershed.

In the *Ehlmann et al.* [2011] model, there is also a brief period of open-system weathering and surface aqueous activity near the Noachian-Hesperian boundary, when the Jezero crater paleolake is inferred to have been active [*Fassett and Head,* 2008a]. It is during this period of surface aqueous activity that *Ehlmann et al.* [2011] also suggest the majority of the Al-rich phyllosilicates (e.g., kaolinite and montmorillonite) identified in ancient Martian terrains formed. In the Nili Fossae region in particular, *Ehlmann et al.* [2009] note the presence of kaolinite in a distinct stratigraphy above Fe/Mg-smectite; however, these occurrences are primarily clustered around the main Nili Fossae trough, and to the south and northeast of the Jezero crater watershed. From our CRISM data analysis, we find Al-rich phyllosilicate-bearing material within the Jezero crater paleolake watershed only in a few small, isolated knobs ($<$~0.1 km^2), which appear to be breccia blocks contained within a geomorphic unit mapped as impact melt from the Hargraves impact crater (Figure 18). It is unclear based on HiRISE images whether the breccia blocks were emplaced coeval with, or prior to the emplacement of the impact melt; however, the blocks appear to show layering (Figure 18b) and bear a similarity to other breccia blocks observed in the altered basement unit (Figure 13e).

Spectra from these breccia blocks exhibit distinct vibrational absorptions at ~1.41, 1.91, and 2.20 μm (Figure 18c). The absorption at ~2.20 μm also has a subtle spectral shoulder at ~2.16 μm. We interpret these vibrational absorptions to be due to a kaolin-group mineral (e.g., kaolinite, dickite, and halloysite). The absorption at ~1.41 μm is caused by the first overtone of OH. The absorption at ~1.91 μm is caused by a combination tone of OH bend and H–O–H stretch from H_2O as a minor component of the kaolinite structure, as a component of the structure of halloysite, and/or from an additional hydrated phase. The absorption at ~2.20 μm is caused by a combination tone of the OH stretch and Al-OH bend in the kaolin-group mineral structure [*Clark et al.,* 1990].

While it is possible that this kaolin-group mineral-bearing outcrop is an example of the regional kaolinite overlying Fe/Mg-smectite stratigraphy [*Ehlmann et al.,* 2009], the morphology and geologic setting is not consistent with this hypothesis. Rather, this outcrop appears to be an isolated breccia block that was altered to a kaolin-group mineral prior to the Hargraves impact, so its origin is difficult to constrain based on available data. Aside from this exposure, we have found no other evidence for Al-rich phyllosilicate minerals in the mapping region.

We hypothesize that the lack of Al-rich phyllosilicates in the Jezero crater watershed area is due to the fluvial activity in this region. During the period of open-system weathering proposed by *Ehlmann et al.* [2011], stratigraphies of Al-rich phyllosilicates overlying Fe/Mg-smectite are produced due to enhanced chemical leaching of Fe/Mg-smectite-rich altered crust. In order to build up such a stratigraphy, the leached soil must remain in place to continually be subject to aqueous chemical weathering; however, if this soil is being actively physically weathered and eroded, new material will continuously be exposed at the surface as old material is transported away, preventing the buildup of intensely leached Al-rich phyllosilicates.

This tradeoff between chemical and physical weathering is a common occurrence on Earth, where the most chemically weathered locations often have the lowest physical weathering, and vice-versa [e.g., *McLennan,* 1993].

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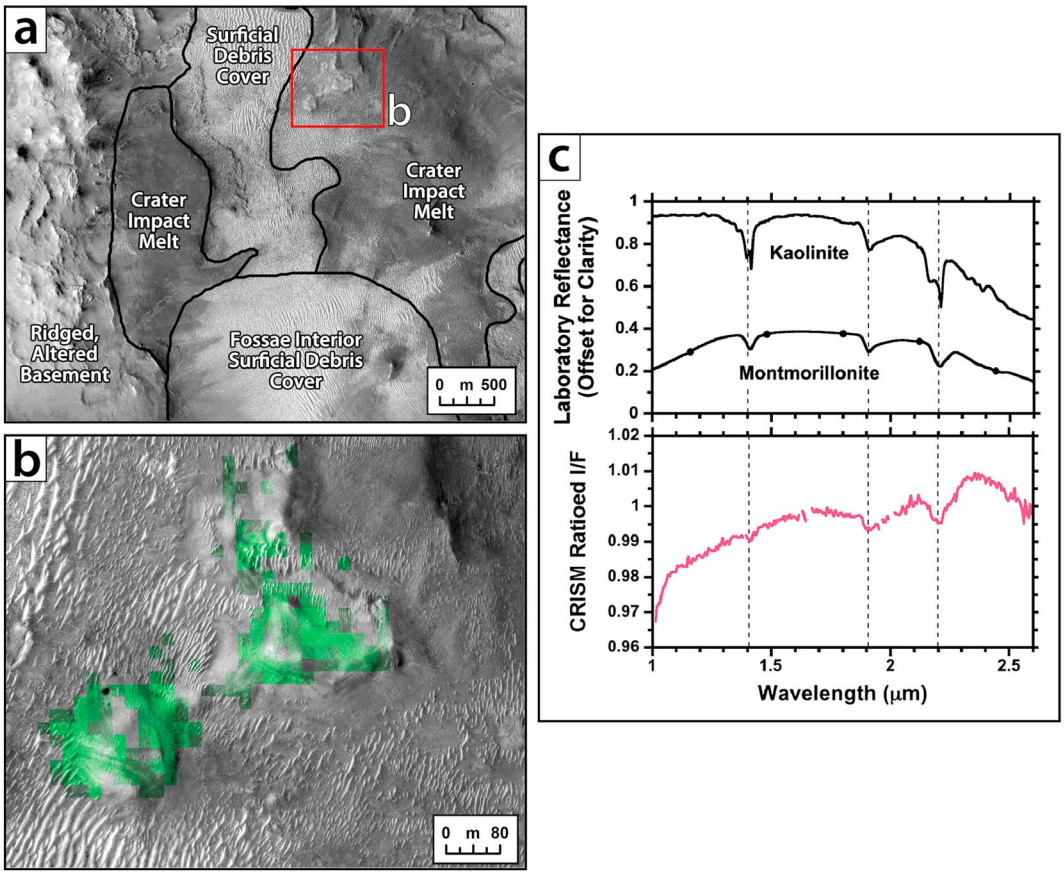

Figure 18. Morphology and spectral signature of the Al-bearing phyllosilicate detection within the Jezero crater paleolake watershed. North is up in all images. (a) Overview and context for the Al-bearing phyllosilicate detection in two layered breccia blocks within a larger expanse of impact melt from the Hargraves impact crater. Location of blowup (Figure 18b) is indicated by labeled red box. Unit contacts are indicated by thick black lines. Background is a portion of HiRISE image PSP_006778_1995 overlain on CTX image P05_002809_1975. (b) Detailed view of the Al-bearing phyllosilicate detection, showing the layered breccia blocks from which the spectrum in Figure 18c is derived. CRISM-derived BD2200 parameter map in green [*Ehlmann et al.,* 2009] from image FRT000097E2 overlain on HiRISE image PSP_006778_1995. Green areas indicate regions consistent with an Al-rich phyllosilicate. (c) CRISM ratioed spectrum of the Al-bearing phyllosilicate detection from the breccia blocks shown in Figures 18a and 18b compared with library spectra of the Al-bearing phyllosilicate minerals kaolinite and montmorillonite. Top plot shows laboratory reflectance spectra of kaolinite (sample LAKA02) and montmorillonite (sample 1292F35) from the CRISM spectral library [*CRISM Science Team*, 2006]. Bottom plot shows a CRISM ratioed spectrum from image FRT000097E2 extracted from the region shown in green in Figure 18b. Dashed lines in both plots are located at ~1.4, 1.91, and 2.2 μm.

Therefore, we hypothesize that physical weathering in the Jezero crater paleolake watershed, for which there is independent evidence in the form of incised valley networks, prevented the buildup of more chemically weathered Al-rich phyllosilicates, explaining the lack of such minerals in comparison to the regional Nili Fossae terrain. We also suggest that the source of the surface water for the period of open-system weathering in the greater Nili Fossae region [*Ehlmann et al.,* 2009, 2011] is the same as that for the period of surface runoff and fluvial activity associated with the formation of the Jezero crater paleolake. This conclusion is consistent with the *Ehlmann et al.* [2011] geologic model for the evolution of early Mars and with the hypothesis that surface fluvial activity on Mars peaked near the Noachian-Hesperian transition [e.g., *Howard et al.,* 2005; *Irwin et al.,* 2005a; *Fassett and Head,* 2008a].

6. Conclusions

Detailed geomorphic mapping in combination with VNIR spectral analyses has revealed a diversity of units in the Jezero crater paleolake system. From this mapping, we have developed a generalized geologic history for the Jezero crater paleolake system (Figures 5 and 16). First, the ancient Noachian crust in the Nili Fossae

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region was altered to a lithology that is spectrally dominated by Fe/Mg-smectite to a depth of at least a few hundred meters [*Mangold et al.,* 2007; *Mustard et al.,* 2007, 2009; *Ehlmann et al.,* 2009]. Following this, the Isidis basin formed in the Middle-to-Early Noachian (~3.96–3.97 Ga) [*Werner,* 2008; *Fassett and Head,* 2011], followed by the formation of the Jezero impact crater. Next, an olivine-rich unit was emplaced in the region, including on the floor of Jezero crater, and subsequently altered to a mixture of Mg-rich carbonate and olivine. A basaltic capping unit, interpreted to be of either volcanic or impact melt origin, which would later erode into the pitted capping unit, was also emplaced upon the altered basement unit at approximately this time.

The next major event was a period of increased surface aqueous activity and runoff in the region, which resulted in the formation of fluvial valleys and valley networks in the Jezero crater watershed, as well as the filling and overtopping of Jezero crater to form the outlet breach. During this time period, two sedimentary fan deposits within Jezero crater [*Fassett and Head,* 2005; *Schon et al.,* 2012] were formed from primarily detrital sediment transported into the basin from the surrounding watershed (Figures 9 and 10).

The two fan deposits show distinct mineralogies based on CRISM hyperspectral reflectance data. The heavily eroded northern fan deposit has a spectral signature consistent with a mixture of olivine and either hydrated Mg-rich carbonate or anhydrous Mg-rich carbonate and an additional hydrated mineral, such as a Fe/Mg-smectite (Figure 10d). The more well-preserved western fan deposit has a spectral signature consistent with Fe/Mg-smectite, such as nontronite or saponite, with minor amounts of Mg-rich carbonate and olivine in isolated exposures (Figure 9d). The results of our source-to-sink analysis indicate that this difference in fan deposit mineralogy is explained by differences in the areal exposure of the major geomorphic units within the two watersheds (Figure 17). This suggests that the aqueous alteration minerals observed in the Jezero crater fan deposits are primarily transported in origin, as opposed to having formed from in situ alteration. Coupled with the observation of sedimentary Fe/Mg-smectite and Mg-rich carbonate within Jezero crater, this conclusion suggests that when the lake was active, the water within the basin was neither highly alkaline nor highly saline and had approximately neutral pH.

Elsewhere in the Nili Fossae region during the period of surface aqueous activity, enhanced chemical weathering from surface runoff and leaching led to the development of a stratigraphy of Al-rich phyllosilicates overlying Fe/Mg-smectite [*Ehlmann et al.,* 2009, 2011]; however, we hypothesize that physical weathering (i.e., fluvial erosion) prevented this stratigraphy from developing in the Jezero crater watershed. The fluvial activity then ceased at ~3.8 Ga [*Fassett and Head,* 2008a], following which the sedimentary fan deposits as well as any lacustrine sediment in Jezero crater were heavily eroded over a time period of ~300–400 Myr. Finally, at ~3.45 Ga, the floor of Jezero crater was resurfaced by a volcanic flow unit [*Goudge et al.,* 2012b] that is <10 m thick along its margins (Figure 11).

Based on our interpreted geologic history of the Jezero crater paleolake system, we conclude that the aqueous alteration minerals observed within the basin are detrital in nature, and that at the study site, the alteration minerals were formed prior to the fluvial activity that fed the paleolake system. This conclusion requires at least two distinct periods of aqueous activity: (1) a period of alteration to form the Fe/Mg-smectite in the altered basement and ridged, altered basement units, and the Mg-rich carbonate in the mottled terrain unit, and (2) a period of fluvial activity and surface runoff to form the Jezero crater paleolake. Our results are consistent with models of early Mars evolution where the majority of crustal aqueous alteration occurred during the Noachian and is followed by a more transient period of surface fluvial activity concentrated at the Noachian-Hesperian boundary [e.g., *Howard et al.,* 2005; *Irwin et al.,* 2005a; *Ehlmann et al.,* 2011; *Fassett and Head,* 2011].

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